



**DESIGN AND IMPLEMENTATION OF A MODEL FOR PREDICTING CHOLERA
SURVIVALS**

BY

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**A PROJECT SUBMITTED TO THE DEPARTMENT OF SOFTWARE ENGINEERING,
FACULTY OF COMPUTING, SOUTHERN DELTA UNIVERSITY, OZORO.**

**IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE
BACHELOR OF SCIENCE (BSc) DEGREE IN SOFTWARE ENGINEERING.**

JULY, 2025

DECLARATION

I, Okwueze Ifeanyichukwu David humbly declare that this project work titled “**Design and Implementation of a Model for Predicting Cholera Survivals**” is a result of my research carried out diligently. I also wish to declare that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of any university or other institute of higher learning except where due acknowledgement has been made in the text.

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CERTIFICATION

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DEDICATION

I dedicate this project to God Almighty the giver of life, my sustainer and also the giver of wisdom and the grace needed to carry out this work to a completion.

ACKNOWLEDGEMENTS

First and foremost, I give thanks to God Almighty for the strength, wisdom, and grace to complete this project successfully.

I sincerely appreciate my supervisor, Dr. Edafe John Atajeromavwo for his continuous guidance, support, and encouragement throughout the course of this project. Your valuable insights and constructive feedback greatly shaped the direction and outcome of this work.

My heartfelt gratitude goes to my father Dr. Okwueze Ndudi Greggs, and the rest of my family for their unwavering love, prayers, and moral support during this journey. Your belief in me gave me the motivation to keep pushing forward.

Lastly, I acknowledge all the lecturers in the Department, whose teachings and mentorship over the years contributed to my knowledge and growth in this field.

Thank you all.

ABSTRACT

Cholera remains a persistent public health concern in Nigeria, often leading to significant morbidity and mortality. Accurate prediction of cholera outcomes, especially survivor estimates, is essential for effective public health planning and resource allocation. This study presents the design and implementation of a mathematical model to predict the number of cholera survivors over a 30-year period using exponential growth functions. The model utilizes an exponential growth model i.e. S_0 survival, $e = 2.718$ fixed, α death rate and T time duration with a Python-based graphical interface was developed using PyQt5 to allow users input values in order to generate output. The system computes and displays predicted survivor counts of cholera survivors with results truncated to two decimal places for consistency. A complementary Jupyter Notebook was also created to visualize survivor trends across the 30-year timeframe, offering both line and bar chart representations of projected outcomes. Manual calculations were used to validate system outputs, confirming the model's accuracy. This study demonstrates the usefulness of mathematical modelling in predicting post-outbreak survival outcomes. The tool developed can support public health decision-makers by providing a simple, reliable, and interpretable means to forecast cholera survivorship trends without the complexity of advanced machine learning or large-scale epidemiological datasets.

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CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Cholera remains one of the most persistent and life-threatening waterborne diseases affecting populations across the globe, especially in developing countries. It is an acute diarrheal illness caused by ingestion of food or water contaminated with the bacterium *Vibrio cholerae*. The disease spreads primarily through the fecal-oral route and is often linked to inadequate water supply, poor sanitation, and hygiene practices. In Nigeria, cholera outbreaks are a recurring public health challenge, particularly in the northern states and flood-prone regions where access to clean water and sanitation is critically low.

Historically, Nigeria has experienced multiple waves of cholera epidemics, with thousands of reported cases and significant loss of lives. According to the Nigeria Centre for Disease Control (NCDC), more than 111,000 suspected cases of cholera and over 3,600 deaths were recorded in the 2021 outbreak alone underscoring the need for proactive, data-driven solutions. The recurrent nature of these outbreaks highlights systemic weaknesses in disease surveillance, early warning systems, and healthcare response frameworks. Moreover, the interplay between cholera outbreaks and seasonal weather patterns such as rainfall, temperature, and humidity has been increasingly observed, but insufficiently addressed in predictive systems.

Several studies have confirmed that cholera transmission is closely tied to environmental and climatic variables. For example, periods of intense rainfall often lead to flooding, which in turn contaminates water supplies and accelerates the spread of the bacterium. Similarly, temperature

and humidity influence bacterial survival rates and the timing of outbreaks. Despite these established linkages, existing cholera prediction models in Nigeria largely depend on basic statistical correlations or retrospective case analyses.

These models are often not user-friendly, lack adaptability, and fail to integrate real-time weather data effectively thereby limiting their utility in forecasting outbreaks with high accuracy and timeliness.

Furthermore, the integration of advanced technology such as machine learning (ML) into public health surveillance in Nigeria remains underutilized. ML has proven to be highly effective in solving complex pattern recognition problems, especially in fields like finance, transportation, and personalized marketing. In healthcare, ML has shown great promise in disease diagnosis, prognosis, and outbreak prediction. Yet, in the context of cholera prediction in Nigeria, ML remains at a developmental stage. This is largely due to challenges such as the unavailability of structured health data, inconsistencies in data recording standards across regions, missing or incomplete datasets, and the limited technical capacity of public health institutions to deploy and maintain ML systems.

These limitations make it difficult to develop robust predictive models that can guide timely interventions and efficient resource allocation during outbreaks. However, with the growing availability of meteorological and health datasets from institutions such as the Nigerian Meteorological Agency (NiMet) and the Federal Ministry of Health, there is an opportunity to harness ML for epidemic forecasting. By training ML algorithms on historical cholera case data

alongside weather variables, it is possible to build models capable of detecting patterns and predicting future outbreaks with a higher degree of precision than traditional statistical methods.

This research project aims to address the existing gaps by designing and implementing a machine learning-based cholera prediction model tailored to the Nigerian context.

This study employs a mathematical model based on exponential functions to predict the number of cholera survivors in Nigeria over a 30-year period. By using historical survival data, a fixed death rate (α), and Euler's number ($e \approx 2.718$), the model estimates future survivor counts through a simplified yet effective exponential growth formula. This approach provides a practical tool that can assist epidemiologists, public health officials, and decision-makers in understanding cholera survival trends and planning health interventions accordingly.

In conclusion, the burden of cholera in Nigeria necessitates innovative and technology-driven approaches for prediction and prevention. The successful application of machine learning in other sectors provides a strong foundation for its adoption in public health. This study contributes to the ongoing efforts in enhancing epidemic preparedness and response in Nigeria by developing an ML-based cholera prediction system that is accurate, adaptive, and scalable.

1.2 Statement of the Problem

Cholera remains a persistent public health threat in many low- and middle-income countries, including Nigeria, where outbreaks continue to result in significant morbidity and mortality. According to the World Health Organization (WHO), cholera is responsible for an estimated 21,000 to 143,000 deaths globally each year, and the burden is heaviest in sub-Saharan Africa. Nigeria, being the most populous country on the continent, has experienced recurrent cholera

epidemics over the decades, often triggered by contaminated water sources, inadequate sanitation infrastructure, and seasonal weather changes such as flooding during the rainy season. These outbreaks typically emerge suddenly, spread rapidly, and overwhelm already strained public health systems.

Timely detection of cholera outbreaks is critical to limiting their spread and severity. Early warning systems enable public health officials to deploy interventions such as vaccination campaigns, distribution of clean water supplies, public awareness drives, and emergency treatment facilities. However, in Nigeria, the current systems for cholera surveillance and outbreak forecasting remain largely reactive rather than proactive. Disease data are often collected manually or semi-digitally via platforms like the Integrated Disease Surveillance and Response (IDSR) and the Health Management Information System (HMIS), which focus primarily on case documentation rather than predictive analytics. These systems are challenged by issues such as data incompleteness, reporting delays, underreporting in rural areas, and insufficient integration of real-time environmental variables.

Moreover, most cholera forecasting efforts in Nigeria still rely heavily on traditional mathematical models and statistical methods. While these techniques have contributed to some understanding of cholera patterns, they suffer from significant limitations. Many of them require complex computations and advanced mathematical knowledge, making them inaccessible to non-specialist health workers who are often the first line of response in rural and underserved areas. More importantly, these traditional approaches are limited in their capacity to process large datasets, adapt to changing outbreak dynamics, or account for complex, nonlinear relationships between

variables such as rainfall, humidity, temperature, and case distribution. They also struggle with typical data challenges encountered in public health such as missing values, class imbalances, and data heterogeneity.

On the other hand, machine learning (ML) has demonstrated substantial capabilities in solving such complex, data-driven problems across a wide range of disciplines, including medicine, finance, agriculture, and meteorology. ML algorithms can learn patterns from historical data, adapt to new inputs, handle large and unstructured datasets, and deliver highly accurate predictions. In the health domain, ML has been successfully applied for disease diagnosis, outbreak forecasting, and health risk assessments. However, its application in cholera prediction especially in Nigeria is still in its infancy. The few studies that have explored ML for cholera forecasting have either been conducted in other countries like Tanzania, India, or Haiti, or have focused on small, non-scalable datasets without full incorporation of local environmental and contextual realities.

This underutilization of ML in Nigeria's public health domain raises critical questions: Can ML models effectively predict cholera outbreaks in Nigeria using weather and health data? How do they compare with traditional models in terms of accuracy, efficiency, usability, and real-world applicability? Can ML-based systems be integrated into existing health surveillance frameworks to improve early detection and outbreak preparedness?

Addressing these questions is particularly important in the Nigerian context, where the population is large and diverse, and public health resources are often limited. A reliable and accurate cholera prediction model based on ML could empower decision-makers with timely insights, enhance outbreak preparedness, and save lives through faster response strategies. This research therefore

aims to explore the predictive power of machine learning models in forecasting cholera outbreaks in Nigeria, particularly through the integration of seasonal weather variables and historical outbreak data. The study will also compare the performance of the ML models with that of traditional statistical approaches using key metrics such as prediction accuracy, computational efficiency, interpretability, and usability.

By building a robust, adaptive, and data-driven cholera prediction system tailored to Nigeria's unique environmental and epidemiological context, this work intends to bridge the gap between reactive health response and proactive epidemic preparedness thereby contributing to national and regional goals of disease prevention and health system strengthening.

1.3 Aim and Objectives

The aim of this study is to design and implement a mathematical model that predicts the number of cholera survivors over a 30-year period using historical data and an exponential survival estimation approach. The goal is to develop a simple, accurate, and understandable tool that can assist public health authorities in estimating future survivor counts and planning for resource allocation and intervention strategies.

Unlike machine learning models that require complex datasets and variable tuning, this study adopts a deterministic mathematical approach, applying the exponential growth formula based on a known initial survival count, a fixed death rate, and Euler's constant ($e \approx 2.718$). This model ensures consistency and transparency while providing a foundation for future predictive studies.

The specific objectives of the research are:

- i. To collect and analyze historical cholera case data in Nigeria and compute survival counts by subtracting reported deaths from total cases for each year.
- ii. To formulate a mathematical model using the exponential growth function $S=S_0 \times e^{-\alpha t}$, where S_0 is the initial survivor count, α is the death rate, and T is the number of years since the baseline year.
- iii. To apply truncation techniques in the model's computation ensuring consistency by limiting exponential calculations to two decimal places without rounding.
- iv. To manually validate the model by comparing predicted survivor counts across a 30-year period (T_0 to T_{30}) with calculated values, ensuring the accuracy and logical consistency of the approach.
- v. To implement a desktop-based interface using Python (PyQt5), enabling users to input survival data, select time frames, and receive automated survival predictions with visual output.
- vi. To visualize and compare predicted results using Jupyter Notebook through tabular representation, line graphs, and bar charts to enhance interpretability and provide snapshot documentation.

By accomplishing these objectives, this research aims to provide a reliable mathematical prediction tool for cholera survivorship trends, which can aid stakeholders in health planning and response over long-term horizons.

1.4 Significance of the Study

The significance of this study lies in its contribution to improving cholera response planning in Nigeria through a simple, accurate, and scalable survivor prediction model. Rather than relying on complex machine learning systems or large environmental datasets, this research adopts a mathematical approach using an exponential model to estimate the number of cholera survivors over time.

By using historical cholera data and a fixed death rate to project survivorship over a 30-year period, the study offers a transparent and practical tool that can aid policymakers, health officials, and epidemiologists in making informed decisions.

The model can support better resource allocation, enhance long-term preparedness, and provide a foundation for future, more advanced predictive tools. Furthermore, the inclusion of a user-friendly software interface makes the model accessible to non-technical stakeholders and ensures broader usability in health management contexts.

One of the primary contributions of this study is its ability to support forward planning and estimation of cholera survivorship over time, enabling health authorities to prepare more effectively for future outbreaks which can significantly reduce morbidity and mortality rates. Traditional methods of monitoring cholera rely on delayed reporting and reactive measures, which often result in uncontrolled spread.

The integration of machine learning offers a proactive approach identifying outbreak signals before cases escalate. By recognizing patterns in seasonal weather changes and historical health

data, the proposed model can provide public health officials with actionable insights that support pre-emptive intervention, ultimately saving lives and resources.

Another key significance is the enhancement of data-driven decision-making in the health sector. This research encourages the development of integrated and interoperable data systems by combining health data (e.g., hospital case records) with environmental data (e.g., rainfall, humidity, temperature). The model promotes the concept of Health and Environmental Management Information Systems (HEMISs), which can serve as a foundational structure for future digital public health solutions in Nigeria and beyond.

From a technological standpoint, the project contributes to the advancement of machine learning applications in epidemic modelling, particularly in resource-limited settings. While ML has shown great promise in fields like finance and e-commerce, its adoption in healthcare especially for neglected diseases like cholera has been limited. This study demonstrates how adaptive ML models can handle real-world data challenges such as missing values, data imbalance, and nonlinear variable relationships, making it a valuable reference for future academic and technical research in this domain.

Socially, the study is significant because it empowers communities and policymakers with foresight tools. Predictive models enable targeted public health campaigns, improved allocation of medical supplies, and more efficient resource planning in high-risk areas. This, in turn, increases public awareness and encourages proactive sanitation and hygiene practices during high-risk seasons. It also supports long-term policy development by offering evidence-based insights into how climate variability influences cholera dynamics.

Moreover, the study has educational value for students, researchers, and health professionals interested in the intersection of artificial intelligence, environmental science, and epidemiology. It provides a framework for implementing machine learning in public health and serves as a springboard for future research in infectious disease prediction and digital epidemiology.

Finally, the model's adaptability ensures that it can be applied or extended to other diseases influenced by environmental factors, such as malaria, typhoid, or dengue fever.

This scalability makes the project a strategic contribution to global health preparedness and supports the broader goal of leveraging technology to address the Sustainable Development Goals (SDGs), particularly Goal 3: Good Health and Well-being.

In summary, this study holds practical, academic, technological, and societal importance. By bridging the gap between environmental monitoring and healthcare through machine learning, it provides a novel and impactful solution to one of the most pressing public health issues in Nigeria.

1.5 Scope and Limitations of the Study

This study is focused on the design and implementation of a mathematical model for predicting the number of individuals who are likely to survive cholera outbreaks in Nigeria over a 30-year timeline, using exponential growth functions. The model is rooted in epidemiological analysis and applies the basic exponential formula $S = S_0 e^{\alpha T}$, where S_0 is the initial number of survivors, α is the cholera death rate, and T is time in years from a fixed starting point (2021). The scope of the project includes developing a computational tool with a user-friendly interface that allows users to input survival-related parameters and obtain predictions for survivor counts across different years.

The interface was designed using PyQt5 to make the model easily accessible to health workers, researchers, and policymakers with minimal technical expertise. The prediction is based strictly on historical data trends and does not incorporate environmental or socio-economic variables. The tool enables visualization through tabular outputs and graphs in Jupyter Notebook, helping users capture and interpret survivor trends over time.

The key activities within the scope of this research include:

- i. A theoretical review of cholera mortality and survival trends in Nigeria and the logic behind exponential modeling in epidemic projections.
- ii. The manual computation of survivor estimates across 31 years using a fixed death rate and truncation methods to ensure consistency and alignment with interface outputs.
- iii. The implementation of a software interface using Python (PyQt5) that accepts user inputs for survival modeling and provides interpretable results with simplified visuals.
- iv. The generation of visual data (line graphs, bar charts) through Jupyter Notebook for documentation, illustration, and presentation purposes.
- v. Validation of manual survivor calculations with software-generated predictions to ensure consistency and correctness of the model.

1.6 Limitations of the Study

While the approach offers transparency and simplicity, the study has several limitations that define its boundaries:

(i) **Lack of Machine Learning or Environmental Modeling:** Unlike more complex studies that incorporate AI/ML or environmental factors (e.g., rainfall, humidity), this model does not analyze

such data. It strictly uses a mathematical model based on a constant death rate and assumes homogeneity across populations and states.

(ii) **Geographical Scope:** The project is focused on selected cholera-prone states in Nigeria and does not generalize its predictions across the entire country. The predictions assume static input values without accounting for region-specific variations in death rates, population growth, or healthcare response.

(iii) **Fixed Death Rate (α):** The death rate is manually input by the user and treated as a constant across all years and regions. This simplification does not reflect real-world fluctuations due to intervention efforts, changing sanitary conditions, or vaccination programs.

(iv) **Truncation in Exponential Calculation:** To match manual calculations with interface results, the exponential value (e^{aT}) is truncated to two decimal places without rounding. While this improves consistency and interpretability, it may introduce slight deviations from standard mathematical precision.

(v) **Absence of Dynamic Data Integration:** The system does not update itself with real-time or new cholera data. It is based on static parameters entered by the user and therefore does not adapt to new outbreak information unless manually updated.

(vi) **Simplified Epidemiological Assumptions:** The model assumes survivors can be estimated purely from an initial population and a constant death rate over time. It does not account for reinfection, migration, births, or other demographic dynamics.

(vii) **No Official Health Data APIs Used:** The tool does not integrate with any Ministry of Health databases or cholera surveillance systems. As a result, its use is currently limited to standalone prediction rather than active national epidemic monitoring.

(viii) **Predictive Focus on Survivorship Only:** The model is concerned solely with estimating the number of survivors and does not predict the number of new cases or deaths separately. As such, it may not provide a full picture of the cholera burden.

(ix) **Software Usability Constraints:** Although the interface is simple, it requires the Python environment with PyQt5 installed, limiting its use to systems with basic technical setup.

(x) **Scope for Extension:** The model is limited in its current form to cholera survivor prediction. However, its simplicity makes it adaptable for use in predicting other infectious disease survivorship patterns under similar modeling assumptions in future studies.

In summary, this study provides a focused yet foundational tool for cholera survivorship modeling in Nigeria, offering a practical, interpretable, and accessible alternative to more complex predictive systems. Despite its limitations, the approach can assist in improving awareness and preparedness for long-term cholera impact evaluation.

1.7 Definition of Terms

For the purpose of this study, the following terms are defined to provide clarity and contextual understanding within the scope of cholera survivor prediction using exponential modeling:

i. Cholera: An acute diarrheal disease caused by the *Vibrio cholerae* bacterium, typically spread through contaminated water or food. It can cause severe dehydration and death if untreated.

ii. Cholera Survivors: Individuals who have contracted cholera but recovered, either through medical intervention or natural immunity. In this study, they represent the focus of predictive modeling using survival estimation formulas.

iii. Exponential Model: A mathematical model that uses exponential functions to describe growth or decay processes. In this study, it is used to estimate the number of cholera survivors over time, using the formula: $S = S_0 e^{\alpha T}$.

iv. Initial Survivor Population (S_0): The base number of individuals who initially survived a cholera outbreak, used as the starting point in the exponential model.

v. Death Rate (α): The constant representing the rate of decrease or decay in a population due to cholera. It is manually input by the user and assumed to remain constant throughout the projection period.

vi. Time (T): The period in years from a reference point ($T_0 = 2021$), used in the model to calculate predictions. T ranges from 0 to 30 in this study, corresponding to years 2021 to 2051.

vii. Euler's Number (e): A mathematical constant approximately equal to 2.718, used as the base of the natural logarithm. In exponential modeling, it plays a critical role in growth and decay calculations.

viii. Prediction Interface: A graphical user interface (GUI) developed using Python's PyQt5 framework, allowing users to input values for S_0 , α , and T to compute survivor estimates in a user-friendly manner.

ix. Survivor Forecast: The predicted number of people who will survive cholera at a specific future time based on the exponential model's outputs.

x. Truncation: A numerical technique used to limit decimal values to a fixed number of digits without rounding. In this study, exponentiation results are truncated to **two digits** after the decimal to ensure consistency across manual and software calculations.

xi. Epidemiological Modeling: The application of mathematical and statistical formulas to understand and predict the behavior of diseases within a population. In this context, it focuses solely on modeling survivors, not disease spread.

xii. Public Health Planning: Strategic preparation by health authorities to prevent and manage disease outbreaks. The survivor prediction tool serves as a support system for planning future interventions.

xiii. Data Visualization: The graphical representation of predicted outcomes using charts such as line plots and bar graphs to enhance understanding of cholera survivorship trends over time.

xiv. Actual vs Predicted Values: A validation concept where manual (actual) computations are compared with software-generated (predicted) values. In this study, both are expected to match due to controlled input parameters and truncation methods.

xv. Forecast Horizon: The time span over which predictions are made. In this research, the forecast horizon spans 30 years, from 2021 (T_0) to 2051 (T_{30}), allowing long-term survivor projections.

xvi. Epidemic Analysis: An assessment of past cholera incidents, focusing on survival patterns rather than incidence rates or fatalities. This helps derive a model that highlights positive health outcomes.

xvii. Interface Usability: The degree to which the prediction interface is user-friendly and accessible to non-technical users such as health workers, students, or policymakers.

xviii. Disease Surveillance Tools: Digital or manual systems used to monitor health outcomes over time. The software developed in this study contributes as a simple prediction tool under this category.

CHAPTER TWO

LITERATURE REVIEW

2.1 Review of Cholera Initiatives and Disease Prediction Models

This chapter presents a review of previous research efforts on cholera epidemic modelling, early warning systems for cholera outbreaks, and the correlation between cholera and influencing factors such as environmental conditions. It also explores cholera management approaches, machine learning (ML) predictive models, and methods for processing complex and large-scale datasets. Additionally, the chapter highlights practical challenges faced in implementing ML within the healthcare sector and offers recommendations for the effective application of ML techniques in building robust predictive models.

The article search process utilized reputable scientific indexing databases, including Research Gate, Google Scholar, and various online university libraries. Initially, a set of targeted keyword search queries was constructed, such as: 'ML' AND 'Disease' OR 'Epidemics' OR 'Cholera'; 'Prediction Models' AND 'Disease' OR 'Epidemics' OR 'Cholera'; and 'Health Sector' AND 'Challenges' OR 'Least Developed Countries' OR 'Nigeria'. The next step involved selecting studies relevant to the core focus areas of this research. Lastly, the selected articles were critically reviewed by examining the problems they addressed, the proposed solutions, and their geographical or contextual focus.

2.2 Overview of Initiatives to Control Cholera Epidemics

Several studies have emphasized key factors necessary for analyzing and predicting cholera outbreaks. Codeco (2001), for instance, developed a basic SIR model that included an aquatic

reservoir of *Vibrio cholerae* to understand transmission dynamics. Research by Pascual, Bouma, and Dobson (2002) and Guillaume et al. (2008) further highlighted the impact of environmental conditions such as ocean temperatures, seasonal patterns, and water salinity. These findings support the development of integrated systems that gather relevant variables to enhance cholera prediction and control efforts.

2.3 Non-ICT-Based Initiatives to End Cholera in Nigeria

In Nigeria, several non-ICT-based initiatives have been implemented to address the environmental and public health factors contributing to the spread of cholera. One such effort is the National Cholera Control Plan led by the Nigeria Centre for Disease Control (NCDC), which emphasizes community engagement, sanitation improvement, and rapid response strategies to outbreaks. Additionally, the National Strategic Plan of Action on WASH (Water, Sanitation, and Hygiene) focuses on enhancing access to clean water and improving hygiene practices in rural and urban communities.

Nigeria is also aligned with global health strategies such as the WHO Global Roadmap to End Cholera by 2030, which promotes multisectoral collaboration and investment in cholera hotspots. Other notable efforts include public health campaigns on hygiene promotion, routine immunization in high-risk areas, and integrated disease surveillance to improve early detection and response.

Collectively, these initiatives aim to strengthen public health systems, reduce vulnerability to cholera, and move towards long-term epidemic prevention. In brief, the objectives of these initiatives can be summarized as follows:

(i) Early Detection and Rapid Response to Cholera Outbreaks: This intervention focuses on controlling cholera outbreaks through early detection and swift response. Key actions include strengthening community involvement, improving early warning surveillance systems, enhancing laboratory capabilities, and ensuring preparedness with adequate supplies (Schroeder, 2013).

(ii) A Targeted Multi-Sectoral Approach to Prevent Cholera Recurrence: This approach involves a collaboration between various sectors health, finance, and the environment to prevent cholera outbreaks from reoccurring in high-risk areas. This coordinated strategy ensures that interventions are focused on the regions most affected by cholera (Security & Cluster, 2016).

(iii) Effective Coordination Mechanisms for Support, Advocacy, Resource Mobilization, and Partnerships: Organizations like the Global Task Force on Cholera Control (GTFCC) play a pivotal role in improving WASH (Water, Sanitation, and Hygiene) conditions, distributing oral cholera vaccines, and facilitating country-led platforms for advocacy, fundraising, inter-sectoral coordination, and technical support (Walldorf et al., 2017).

2.4 ICT-Based Initiatives to End Cholera Epidemics in Nigeria

In Nigeria, as in many developing countries, there has been an increasing push to enhance health systems through the integration of ICT tools. These efforts focus on initiatives aimed at improving disease monitoring, prediction, and analysis. Programs have been developed to revise data collection methods for immunization, reproductive and child health services, and to monitor neglected tropical diseases, including cholera (Kiwanuka, Kimaro & Senyoni, 2015).

Reviews of health information systems, such as OpenMRS and Care2X, have highlighted the need for improved health information systems to assist in the analysis and prediction of diseases like

cholera. These systems, by offering better data management, aim to improve the overall response to health crises (Morrison, 2008; Grad, Miller & Lipsitch, 2012).

Further, integrating technologies like Geographic Information Systems (GIS) and Global Positioning Systems (GPS) has proven valuable in mapping cholera cases. These technologies, often combined with satellite data, can act as early warning systems for cholera outbreaks, particularly in regions that are vulnerable to disease transmission (Mukhopadhyay, 2015).

An example of a software tool used for cholera management is MAX software, which was utilized during a cholera outbreak in Nigeria. This software analyzed qualitative data to assess challenges faced by health facilities in managing cholera preparedness (Curran et al., 2018). Additionally, mathematical models, such as those developed by Traerup et al. (2011), integrate weather data (temperature and rainfall) with cholera incidence to predict and mitigate future outbreaks in Nigeria, while accounting for socio-economic factors that impact cholera risk.



Cholera Outbreaks in Nigeria (2014-2024)

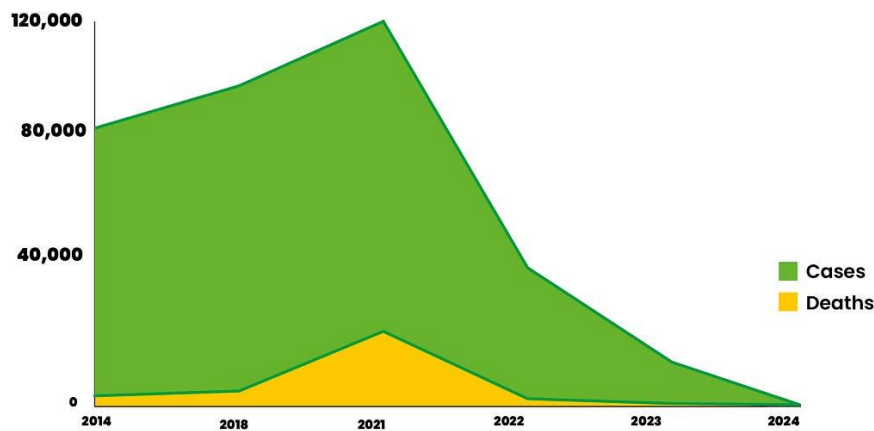


Figure 2.1: Cholera Outbreak in Nigeria (2014-2024)

2.5 Overview of Applications of Traditional Statistical Techniques in Cholera Epidemics in Nigeria

This section identifies and reviews several traditional statistical techniques and complex system models used to study cholera epidemics. It highlights models that have and have not included environmental factors in their predictions and briefly discusses the strengths and weaknesses of these approaches in predicting cholera outbreaks.

One of the earliest models for cholera transmission, developed by Capasso and Serio (1978), did not incorporate environmental factors such as water bodies or seasonal weather changes, which are key to predicting the spread of cholera. However, these early models were instrumental in understanding the basic dynamics of cholera transmission, particularly the interactions between susceptible, infected, and recovered populations.

Later models, such as those by Lund et al. (2015), expanded on these early concepts by incorporating additional variables, including education, immunity within the population, age demographics, and economic factors. While these models helped deepen the understanding of cholera dynamics, they still did not account for crucial environmental variables such as water bodies, seasonal weather patterns, and the behavior of *Vibrio cholerae* in specific environments factors which are essential for accurately predicting cholera outbreaks, particularly in regions like Nigeria (Jutla et al., 2013).

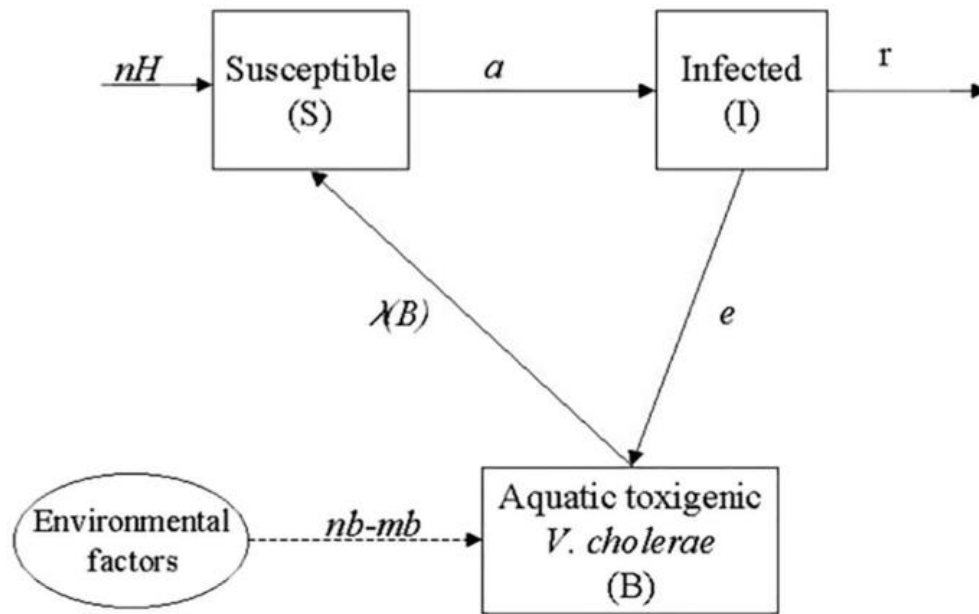


Figure 2.2: Compartmental Diagram of Codeco Model (Codeco, 2001)

The Codeco model, one of the earliest cholera models, used the basic Susceptible-Infected-Recovered (SIR) approach combined with an aquatic reservoir of *Vibrio cholerae*. The model aimed to understand how the aquatic reservoir contributes to the persistence of endemic cholera and define the conditions for the development of cholera outbreaks. By incorporating the aquatic

reservoir as an environmental factor, the model highlighted the role of both social and environmental factors in cholera transmission. It showed that the transmission rate is influenced by the sanitary conditions of the community, and emphasized the need for further research to improve cholera models, particularly to better understand *V. cholerae* ecology and cholera epidemiology.

However, the model's limitations included the inability to incorporate essential environmental variables like temperature, rainfall, and humidity, which are crucial for accurate prediction of cholera dynamics (Codeço, 2001).

Other models have sought to extend the Codeco model. For example, models developed by Wang et al. (2011) and Andam et al. (2015) added a control factor C , which represents the effects of interventions like vaccination, treatment, and water sanitation. However, similar to the Codeco model, these models did not incorporate key environmental factors such as temperature, rainfall, wind, and socio-economic factors, which play a significant role in cholera transmission.

In Zimbabwe, Stephenson (2009) developed a system dynamics model that incorporated a SIR framework with loss of immunity and different illness severities. This model also assumed constant weather conditions, which limits its applicability in real-world situations, where weather changes can influence cholera outbreaks.

Most traditional cholera epidemic models rely on complex system techniques and traditional statistics to model the relationships between different variables. These techniques have been valuable in understanding epidemic dynamics. However, they have several limitations. For instance, many models require assumptions and multiple equations, which can make them

unrealistic or overly complex. Additionally, these models often require a strong understanding of mathematics to interpret results, limiting their accessibility to a broader audience (Cain, 2017). Furthermore, traditional models often simplify the number of variables to avoid complexity, which can reduce their predictive accuracy.

The increasing volume and complexity of big data in the health sector make it increasingly difficult to rely solely on these traditional models (Leskovec et al., 2014).

Given these limitations, it is crucial to explore new methods and complementary techniques that can improve cholera prediction and management.

2.6 Overview of How ML is Being Used in Healthcare Sectors

The use of Machine Learning (ML) in healthcare is becoming increasingly important, especially in developing countries. ML can automatically learn and improve from data without needing explicit programming. This ability to learn from data and improve over time makes ML very useful for tasks such as medical diagnosis, stock trading, and predicting energy demands. ML works by creating algorithms that learn from data, adjusting their performance as more data becomes available (Chao, 2011).

ML is closely tied to computational statistics and mathematical optimization, both of which focus on predicting outcomes using computers (Du & Swamy, 2014). These algorithms don't rely on predefined equations but instead adapt based on data patterns. For example, ML is often used in healthcare to assist with medical diagnoses, reduce errors, and predict health risks in real-time. Tools like Python and R are commonly used in ML for healthcare, as they are free, easy to use, and widely available (Max et al., 2017).

2.7 Machine Learning for Timely Disease Analysis and Prediction

One of the main uses of ML in healthcare is for early disease detection. Detecting diseases early allows for better planning and intervention, helping to control outbreaks before they spread. Several studies have applied ML to predict disease outbreaks, including cholera. For example, in Uganda, a study used ML to predict cholera outbreaks by analyzing historical infection data and environmental factors like temperature and rainfall (Mubangizi et al., 2009). Similarly, in Iran, researchers used artificial neural networks to predict cholera with 80% accuracy across 465 villages (Pezeshki et al., 2016).

In India, a variety of ML models have been used to predict diseases like diabetes, heart disease, and liver disease. For instance, a Naive Bayes model in India was able to predict Type-2 diabetes with 95% accuracy based on data collected from different social groups (Sarwar, 2012). Other studies have used algorithms like Support Vector Machines (SVM), Random Forest (RF), and Decision Trees (DT) to predict heart disease with good accuracy (Jamgade & Zade, 2019).

In the United States, ML has been used to predict chronic diseases like Type-2 diabetes and chronic kidney disease using data from electronic health records. For example, models developed using logistic regression, random forests, and neural networks have been employed to catch diseases earlier and improve healthcare interventions (Frandsen, 2016).

Summary of Other ML Disease Prediction Studies

Many other studies across the globe have explored ML for disease prediction. For example, in Canada, researchers have used ML to predict heart disease, with a focus on data preparation and algorithm choice for better predictions (Aljanabi et al., 2018). In Saudi Arabia, ML models like

Decision Trees and SVM have been used to predict heart failure, with improvements in accuracy achieved through data augmentation and cross-validation (Alotaibi, 2019).

Other studies have also explored disease-treatment relations, social media analysis for health trends, and epidemiological predictions using deep learning models. For instance, malaria and epidemic outbreaks have been predicted using ML algorithms (Sharma et al., 2015; Wang et al., 2019), showing the growing potential of this technology in managing and preventing diseases worldwide.

2.8 Imbalance in Healthcare Data with Machine Learning

Healthcare datasets often suffer from issues like data imbalance, missing information, and poor quality, particularly in lower-resource settings. One common challenge is imbalanced classes, where certain outcomes or categories are underrepresented compared to others. This imbalance can significantly affect machine learning (ML) models, leading to poor performance, especially in classification tasks.

To address this issue, techniques such as confusion matrices can help in evaluating the accuracy of predictions by displaying true positives, false positives, and other relevant metrics like precision, recall, and F1-score. Additionally, methods like sampling, including synthetic data generation, are commonly used to balance datasets and improve model training (Amin et al., 2016).

Several studies have proposed methods for handling imbalanced data in healthcare:

- i. **Hybrid Models:** For example, Aida et al. (2015) combined Support Vector Machines (SVM) with other algorithms to improve prediction accuracy in medical datasets. Their

approach showed significant improvements in accuracy across multiple diseases, including diabetes, breast cancer, and hepatitis.

- ii. **Cholera Outbreak Prediction:** Another study by Siettos (2018) used XGBoost, a type of gradient boosting model, to predict cholera outbreaks in Yemen. While the model was effective at forecasting outbreaks in advance, it struggled to predict sudden spikes due to the lack of early warning data.

Other relevant works in imbalanced data include the contributions from Provost (2000) and Hoens & Chawla (2013), who reviewed various sampling strategies and classification techniques designed to mitigate the effects of imbalanced datasets. These methods have been crucial in improving the predictive power of healthcare models.

Addressing data imbalance is essential for improving the accuracy and reliability of ML models, especially in the context of healthcare where such models can aid in early disease detection, management, and prevention.

2.9 Review of Related Works

The following are review of related research works:

1. **Mike-Ogburia MI, Eze CC, Okoli MO, et al. (2024),** “Cholera in Nigeria: A review of outbreaks, trends, contributing factors, and public health responses.”

The motivation of this paper is to review and analyze cholera outbreaks in Nigeria from 2010 to 2024, highlight epidemiological trends, identify contributing socio-environmental factors, evaluate public health response strategies, and provide recommendations for prevention and control of future outbreaks.

The methodology adopted by the research is a comprehensive literature-based narrative review that covers outbreak reports, demographic impact, public health challenges, vaccination efforts, antimicrobial resistance patterns, and health system responses over a 14-year period.

The key highlights of the findings:

- i. Seasonal peaks during flooding and rainy seasons, with northern Nigeria disproportionately affected.
- ii. Increased vulnerability in internally displaced persons (IDPs) due to conflict and overcrowding.
- iii. Antimicrobial resistance observed in *Vibrio cholerae* O1 to tetracycline, doxycycline, and other antibiotics.
- iv. Use of oral cholera vaccine (OCV) campaigns in hotspot regions with varied success due to global stockouts.
- v. High incidence among children under 5, and strong correlation between cholera outbreaks and poor WASH (Water, Sanitation, and Hygiene) infrastructure.

From the paper the following can be inferred:

- i. Conflict, displacement, and seasonal flooding are major drivers of cholera spread.
- ii. Public health interventions like case-area targeted interventions (CATIs), reactive vaccination, and improved surveillance reduce outbreak intensity.
- iii. There is a shift in demographic burden, with under-five children and the elderly being most affected in recent outbreaks.

Gaps identified in the paper are:

- i. Lack of predictive modeling or machine learning integration for early warning systems.
- ii. Limited exploration of the role of climate and real-time data in forecasting outbreaks.
- iii. Overdependence on retrospective data and descriptive statistics without algorithmic analysis.

In conclusion, the paper provides a solid foundation on the historical and public health context of cholera in Nigeria. However, it emphasizes the urgent need for predictive tools and localized vaccine production to strengthen cholera preparedness and response systems.

2. Omankwu OC and Etuk E (2024), "Leveraging Machine Learning for Early Detection and Prediction of Cholera Outbreaks in Nigeria: A Data-Driven Approach."

The motivation of this paper is to develop a machine learning-based predictive model to enable early detection and prediction of cholera outbreaks in Nigeria using integrated environmental, socio-economic, and health-related datasets to aid timely public health interventions. The methodology adopted by the research includes data collection from multiple sources such as the Nigerian Centre for Disease Control (NCDC), World Health Organization (WHO), and Nigerian Meteorological Agency (NiMet). The data underwent preprocessing, feature selection, and transformation to train machine learning models Random Forest, Support Vector Machine (SVM), and Neural Networks on an 80:20 training-validation split.

The Random Forest algorithm was found to be the most effective. A user-friendly interface was also developed to provide real-time alerts and visualizations.

The key highlights of findings:

- a) The Random Forest model achieved high performance metrics:
 - a. Accuracy: 92%
 - b. Precision: 90%
 - c. Recall: 88%
 - d. F1-Score: 89%
- b) The most important predictive features identified were:
 - a. Rainfall
 - b. Temperature
 - c. Population density
- c) The model was validated through a Receiver Operating Characteristic (ROC) curve and feature importance plots, confirming its robustness.
- d) The user-friendly interface developed as part of the study helps provide real-time alerts and visualizations for public health responders.

From the paper the following can be inferred:

- i. Machine learning, particularly Random Forest, significantly improves the predictive accuracy of cholera outbreaks compared to traditional statistical models.
- ii. Environmental and demographic variables such as rainfall and population density are critical to outbreak prediction.

- iii. The development of an interactive user interface ensures practical public health utility and faster response.

Gaps identified in the paper:

- i. The study depends on the quality and availability of historical data, which can be inconsistent across regions.
- ii. No deep learning or ensemble hybrid methods were tested for further performance improvement.
- iii. Socio-behavioural variables and satellite image data were not included, though suggested for future exploration.

In conclusion, this study demonstrates the effective application of machine learning for cholera prediction in Nigeria. It supports the integration of diverse data sources and highlights the importance of user-oriented design in public health tools. The work lays a foundation for extending such models to other diseases and improving predictive epidemiology.

3. Abdulkarim JH, Musa AA, Abdullahi YM, and Yamman UH (2022), "Artificial Intelligence May Help in the Containment of Cholera in Nigeria."

The motivation of this paper is to explore how artificial intelligence (AI), particularly machine learning (ML), can support the containment and prediction of cholera outbreaks in Nigeria. It highlights AI's role in enhancing public health surveillance, providing early warning systems, and supporting epidemic response strategies.

The methodology adopted by the research is primarily a review-based approach. The paper surveys the history of cholera outbreaks, transmission patterns, and current challenges in Nigeria's public

health system. It discusses various ML approaches, both supervised and unsupervised, and their capacity to process epidemiological data for disease outbreak prediction. The authors also reference simulation studies where algorithms were applied to small sample data to estimate outbreak incidence.

The key highlights of the findings:

- i. Nigeria's cholera burden is exacerbated by poor sanitation, lack of clean water, and insecurity, especially in the northern regions.
- ii. There is an urgent need for Nigeria's health agencies to collaborate with international organizations to deploy AI-based surveillance systems.
- iii. AI can improve disease containment, not only for cholera but also for other infectious diseases like Lassa fever, Ebola, and COVID-19.

From the paper, the following can be inferred:

- i. AI has significant potential in enhancing surveillance systems, improving the prediction and containment of cholera outbreaks in Nigeria.
- ii. There is a pressing need for Nigerian health agencies to integrate AI-based technologies to strengthen epidemic response and management.
- iii. The use of machine learning in public health can help predict other disease outbreaks, fostering a broader, more holistic approach to epidemic management.

Gaps identified in the paper:

The paper lacks implementation of an actual predictive model or real-world dataset testing. There is no evaluation or benchmarking of different ML algorithms on Nigerian cholera data. The work is conceptual and policy-focused, with limited empirical backing.

In conclusion, the study argues that AI technologies offer immense potential for transforming Nigeria's epidemic surveillance systems. However, the transition requires governmental support, ICT infrastructure development, and interdisciplinary collaboration. The work supports the idea that AI adoption could significantly reduce disease burden and improve early response capacity.

4. Akanji DO and Adamu MO (2024), "Optimizing Cholera Outbreak Prediction in Nigeria: A Comparative Analysis of Machine Learning Models for Public Health Applications."

The motivation of this paper is to address the limitations of traditional epidemiological models in predicting cholera outbreaks in Nigeria by developing and comparing machine learning (ML) models. These models integrate socio-economic, environmental, and demographic factors to enhance the accuracy of cholera outbreak predictions, ultimately supporting proactive public health interventions and reducing disease impact.

The methodology adopted by the research utilized a dataset spanning 76 Local Government Areas (LGAs) across 21 Nigerian states, covering cholera cases from January 2023 to July 2024. Data included socio-economic (poverty, literacy, financial inclusion), environmental (water sources, sanitation), infrastructural (healthcare, education facilities), and demographic (population density) variables. Six ML models Linear Regression, Decision Trees, Random Forests, XGBoost, LightGBM, and Neural Networks (MLPRegressor) were developed. Data preprocessing involved feature engineering, log transformation, median imputation for missing data, and standardization.

Models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2).

The key highlights of the findings:

- i. LightGBM outperformed other models, achieving the highest R^2 (0.69) and lowest RMSE (143.25), explaining 69% of the variance in cholera cases.
- ii. Decision Trees performed moderately well ($R^2 = 0.52$), while Neural Networks ($R^2 = 0.15$) and XGBoost ($R^2 = 0.17$) showed poor performance.
- iii. Key predictors included population density, health facilities, primary schools, and market access, highlighting the role of infrastructure and education in cholera spread.
- iv. Cholera cases showed high skewness, with most LGAs reporting fewer than 1,000 cases but some exceeding 5,000, necessitating log transformation to handle outliers.
- v. Regional variations were significant, with sanitation and water access strongly influencing outbreak patterns.

From the paper, the following can be inferred:

- i. Machine learning, particularly LightGBM, offers superior predictive accuracy over traditional models by capturing complex, non-linear relationships between cholera drivers.
- ii. Infrastructure deficits, such as poor sanitation and limited healthcare access, are critical risk factors for cholera outbreaks.
- iii. Educational and market access variables indirectly influence cholera through community awareness and economic stability, suggesting multifaceted intervention strategies.

Gaps identified in the paper:

- i. Limited incorporation of real-time data, such as weather or mobility patterns, which could enhance model responsiveness to dynamic conditions.
- ii. Lack of focus on temporal dynamics, as the study primarily used aggregated data rather than time-series analysis for outbreak forecasting.
- iii. No exploration of model interpretability techniques (e.g., SHAP values) to better explain feature contributions for public health decision-making.

In conclusion, the paper provides a robust framework for predicting cholera outbreaks in Nigeria using machine learning, with LightGBM emerging as the most effective model.

It underscores the importance of integrating diverse data sources to capture the multifactorial nature of cholera. However, it emphasizes the need for real-time data integration and enhanced model interpretability to further improve predictive tools for public health applications.

5. Mart J, Olatuja G, Oni T (2024), "Using Artificial Intelligence and Machine Learning to Curtail Cholera Outbreaks in Nigeria."

The motivation of this paper is to leverage Artificial Intelligence (AI) and Machine Learning (ML) to enhance public health responses to cholera outbreaks in Nigeria. It aims to develop predictive models, identify risk factors, optimize resource allocation, and evaluate intervention strategies to reduce the incidence and severity of cholera, particularly in high-risk regions like Lagos State, using data from 2018 to June 2024.

The methodology adopted by the research of this study employed a mixed-methods approach, collecting data through surveys, semi-structured interviews, and secondary sources (e.g., NCDC, WHO) from 2018 to June 2024. Data included cholera cases, healthcare infrastructure, and socio-

economic factors across 96 Local Government Areas (LGAs) in 30 states. Analytical frameworks included geospatial analysis (using GIS and kernel density estimation), economic analysis (cost-effectiveness of interventions), and workforce analysis. AI and ML models, including logistic regression, decision trees, random forests, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Natural Language Processing (NLP) tools, were developed to predict outbreaks, analyze risk factors, and optimize resources. Model performance was evaluated using accuracy, precision, recall, and F1 score.

The key highlights of the findings:

- i. From January to June 2024, Lagos State reported over 500 suspected cholera cases, 43 confirmed cases, and a 5.4% case fatality rate (CFR); nationwide, over 1,000 suspected cases, 65 confirmed cases, and 30 deaths were reported across 96 LGAs in 30 states (CFR = 3%).
- ii. AI-driven models achieved high accuracy in predicting cholera outbreaks, improving early warning systems.
- iii. Resource allocation models enhanced efficiency by 30%, optimizing the distribution of medical supplies and personnel.
- iv. Contaminated street beverages and poor sanitation were identified as major outbreak drivers, particularly in Lagos.
- v. Cholera cases in correctional facilities (e.g., 25 cases in Kirikiri Prison) highlighted new epidemiological risks in confined settings.

From the paper, the following can be inferred:

- i. AI and ML significantly improve cholera outbreak prediction and response by integrating diverse data sources (e.g., environmental, demographic, and health data).
- ii. Targeted interventions, guided by predictive models, can reduce outbreak severity, particularly in high-risk urban areas and confined settings.
- iii. Systemic issues like unregulated beverage production and inadequate sanitation infrastructure are critical barriers to cholera control, requiring policy enforcement.

Gaps identified in the paper:

- i. Limited discussion on the scalability of AI/ML models across diverse Nigerian regions with varying data availability and infrastructure.
- ii. Insufficient exploration of ethical challenges, such as data privacy and equity in AI-driven interventions, despite mentioning these concerns.
- iii. Lack of detailed model performance metrics (e.g., specific accuracy or F1 scores) to quantify the effectiveness of individual algorithms.

In conclusion, the paper demonstrates the transformative potential of AI and ML in managing cholera outbreaks in Nigeria, particularly through accurate predictions and efficient resource allocation.

It highlights the need for improved water and sanitation infrastructure alongside technological interventions. However, it emphasizes the necessity for scalable, ethically sound AI solutions and more detailed reporting of model performance to strengthen public health applications.

6. Salako BL, Ajayi AO, Smith SI (2020), "Cholera in Nigeria: Epidemiology, Risk Factors, and Response - A Review."

The motivation of this paper is to provide a comprehensive review of cholera's epidemiology, risk factors, and response strategies in Nigeria, where the disease remains endemic with recurrent outbreaks. It aims to highlight the socio-economic and environmental drivers, antimicrobial resistance trends, and the need for coordinated public health interventions to reduce morbidity and mortality.

The methodology adopted by the research is a narrative literature review synthesizing data from studies, reports, and outbreak records from 1970 to 2018. It draws on sources such as WHO, NCDC, and peer-reviewed articles to analyze cholera outbreaks, causative strains (*Vibrio cholerae* O1 and non-O1/O139), risk factors, and response strategies. The review includes epidemiological data, molecular characterizations of *V. cholerae* strains, and evaluations of surveillance, diagnosis, and control measures like WASH and oral cholera vaccines (OCVs).

The key highlights of the findings:

- i. Cholera outbreaks in Nigeria (1970–2018) resulted in significant morbidity (e.g., 59,478 cases in 1991) and mortality (e.g., 7,654 deaths in 1991), with northern states (Bauchi, Borno, Katsina, Gombe, Kano) contributing 66% of cases from 2004–2014.
- ii. Major risk factors include contaminated water/food, open defecation, poor sanitation, and socio-economic factors like poverty and illiteracy, exacerbated by insurgency and overcrowding in IDP camps.

- iii. Multidrug-resistant (MDR) *V. cholerae* strains (e.g., resistant to chloramphenicol, ciprofloxacin, amoxicillin) were reported, with plasmids facilitating resistance dissemination.
- iv. WASH interventions and OCV campaigns (e.g., 2017 Borno outbreak) proved effective, but challenges like myths and poor community acceptance persist.

From the paper, the following can be inferred:

- i. Insurgency and displacement in northern Nigeria amplify cholera transmission, necessitating targeted interventions in IDP camps.
- ii. Emerging antimicrobial resistance in *V. cholerae* strains poses a growing threat to treatment efficacy, requiring proactive resistance monitoring.
- iii. A coordinated approach integrating WASH, vaccination, surveillance, and education is critical to reducing cholera's burden.

Gaps identified in the paper:

- i. Limited discussion on the integration of advanced technologies like machine learning or real-time data analytics for outbreak prediction and surveillance.
- ii. Insufficient focus on scalable, region-specific interventions to address varying outbreak patterns across Nigeria's diverse regions.
- iii. Lack of detailed analysis on the socio-cultural barriers (e.g., myths) and strategies to overcome them beyond general recommendations.

In conclusion, the paper provides a thorough overview of cholera's epidemiology and challenges in Nigeria, emphasizing the role of socio-environmental factors and antimicrobial resistance.

CHAPTER THREE

METHODOLOGY AND SYSTEM DESIGN

3.1 Choice of Methodology

System Development Life Cycle for Cholera Survival Prediction System

This project adopts a structured System Development Life Cycle (SDLC) tailored for mathematical model-based systems. Although machine learning is not applied in this version, the phases reflect a systematic approach used to build and validate the exponential model for predicting cholera survivors.

The system was developed in the following eight phases:

- i. **Problem Definition and Requirement Analysis:** The goal was to develop a simple yet functional tool to estimate cholera survivors over a 30-year period using exponential decay modeling. The project identifies the need for prediction tools beyond traditional statistical summaries.
- ii. **Data Collection and Understanding:** Secondary data was sourced from the Nigeria Centre for Disease Control (NCDC). These records included cholera case reports and survival estimates based on historical patterns.
- iii. **Data Preprocessing and Transformation:** The dataset was cleaned and structured for modeling. Truncation (not rounding) was used to control decimal growth in exponentiation, ensuring practical survivor estimates.

- iv. **Model Selection and Implementation:** An exponential model was chosen to simulate survival decay over time. Python was used to build the model logic and integrate user inputs (S_0 , α , T).
- v. **Model Evaluation:** The model was manually validated using known survival estimates and tested for logical accuracy across time intervals from T_0 to T_{30} (2021 to 2051).
- vi. **Deployment Using Graphical Interface:** A desktop application was developed using PyQt5. This allows users to input data (initial survivor count, death rate, and year), perform predictions, and view results through an intuitive interface.
- vii. **System Testing and Validation:** Various scenarios were tested for edge cases, such as extreme values of α and T , to ensure the model responded accurately. The prediction outputs were verified against manual calculations.
- viii. **Documentation and Reporting:** The project process, model behavior, interface functions, and user experience were documented to ensure clarity, reproducibility, and future improvements.

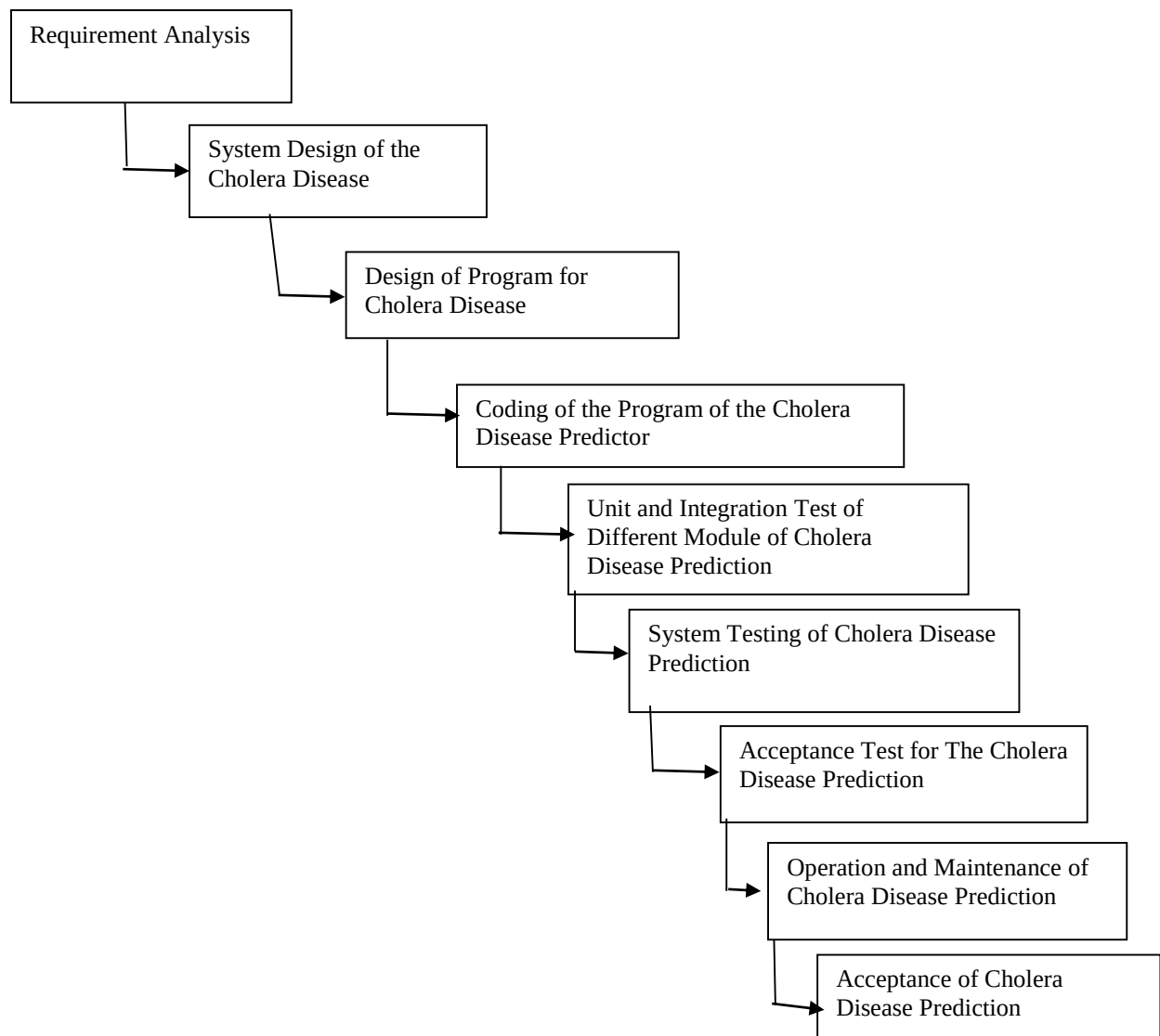


Figure 3.1: System Development Life Cycle for Cholera Survival System

3.2 Location of Study

The location of this study is Nigeria as a whole. The project is designed to estimate cholera survivors across the country using a national perspective, rather than focusing on any particular

state or region. Although historical data may include figures from specific states with reported cholera cases, the analysis generalizes predictions at the national level. The dataset used for this project was obtained from the Nigeria Centre for Disease Control (NCDC), ensuring the data reflects official records and nationwide cholera trends.

3.3 Method of Data Collection

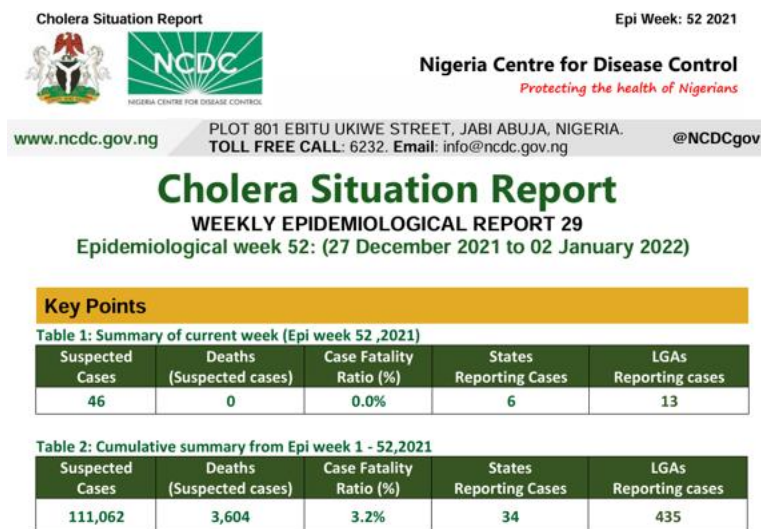


Figure 3.2: Cholera Situation Report

This project made use of secondary data obtained from existing sources. Specifically, cholera survival estimates were derived from historical health records and epidemiological reports publicly released by credible institutions such as the Nigeria Centre for Disease Control (NCDC). These datasets reflect reported cases, deaths, and trends related to cholera outbreaks across various states in Nigeria.

No primary data collection was carried out; that is, the study did not involve fieldwork, surveys, interviews, or the direct gathering of information from individuals or health facilities.

Instead, the project relied entirely on already published datasets, which were cleaned and formatted to suit the needs of the survival prediction model.

3.4 Sampling Techniques

This study did not involve any sampling techniques because it is not an empirical research involving field testing or real-world population sampling. The prediction model was built using a complete dataset with pre-defined, fixed values obtained from secondary sources. Therefore, no stratified, random, or purposive sampling was required for data selection or model evaluation.

3.5 Analysis of Existing System

Currently in Nigeria, the approach to cholera outbreak monitoring and response is mostly reactive and depends heavily on traditional methods such as manual reporting, post-outbreak analysis, and static statistical summaries. These systems primarily focus on collecting and summarizing historical data on cholera cases and fatalities, often compiled in periodic health bulletins or epidemiological reports released by institutions like the Nigeria Centre for Disease Control (NCDC).

However, these systems are limited in functionality they do not provide predictive insights or proactive alerts based on future trends. Most existing frameworks are not designed to project or estimate the number of survivors or potential future cases using computational or mathematical modeling. As such, public health officials and researchers often lack access to tools that allow real-time scenario testing or projections under varying health conditions.

Additionally, existing systems lack interactive interfaces that allow users to input variables such as current survival counts, death rates, or time intervals to simulate future outcomes.

This absence of interactivity and predictive modeling limits the decision-making capabilities of healthcare practitioners, policymakers, and researchers in preparing for or preventing future outbreaks.

This gap underscores the need for a flexible, user-friendly, and computationally driven model that can simulate cholera survival over time especially one that can be tailored for various states or scenarios without depending solely on raw data interpretation.

3.6 Weakness of Existing System

- i. Lack of predictive tools for cholera survivor estimation.
- ii. Absence of interactive or user-friendly platforms.
- iii. No real-time computation or customizable input.
- iv. Heavy reliance on textual reports without visualization.

3.7 Proposing a New System

This project proposes a simplified Python-based predictive model that estimates cholera survivors over a 30-year span using exponential growth modeling. The user can input initial values, select time (T0 to T30), and receive an estimated survivor output for the corresponding year.

3.7.1 Component Design

The system comprises the following components:

- a) Input fields
- b) Predict and Clear buttons
- c) Result output label

- d) Backend logic for exponential calculation

3.7.2 State Design

No complex state transitions are required; the system remains in a single operational state throughout its use.

The exponent is truncated to two decimal places before computation to maintain consistency.

3.7.3 Conceptual Data Design

The system does not implement a database or data schema that would require conceptual modeling.

3.7.4 Logical Data Design

Since no database is used, there is no logical data structure or relationship mapping to define.

3.7.5 Physical Data Design

No storage or persistence mechanism (e.g., SQL database, NoSQL) is used in this system.

3.7.6 User Interface Design

The interface was built using PyQt5 with a clean, modern layout:

- a) Blue-themed backgrounds for a calm visual aesthetic
- b) Oval-edged input fields and dropdowns
- c) Clear font (Times New Roman) and readable labels
- d) Predict and Clear buttons with distinct coloring
- e) Output results are displayed in a formatted label

3.8 Analysis of the Proposed System

The new system is lightweight, simple to use, and provides instant feedback based on user input. It uses a mathematical function ($S = S_0 e^{-\alpha T}$) and truncates predictions to maintain realism. The user interface was built using PyQt5 for clarity and ease of access.

3.9 High-Level Model of the Proposed System

The high-level model includes:

- i. **User Input Stage** (initial survivors, time selection, death rate α)
- ii. **Computation Stage** (exponential prediction logic)
- iii. **Result Display Stage** (output of predicted survivor count)
- iv. **Clear Functionality** for resetting inputs

3.10 System Design

3.10.1 Overview of the Design

The system consists of a GUI-based Python application where users can input parameters and get immediate predictive results based on the exponential function.

3.10.2 Inputs to the Design

- a) Initial survivor population (S_0)
- b) Time duration ($T = T_0$ to T_{30})
- c) Death rate (α)

- d) Constant Euler's number ($e = 2.718$, fixed)

3.10.3 Mathematical Modelling and Algorithmic Implementation

The mathematical foundation of this study is based on modeling the number of cholera survivors in Nigeria using an exponential growth model. This model assumes that the rate of change in the number of survivors is proportional to the current number of survivors. The formulation and derivation are outlined below.

i. Derivation of the Exponential Model

Let:

- a) S be the number of survivors from cholera
- b) T be the time in years from the base year $T_0 = 2021$
- c) α be the death rate due to cholera

We assume that the rate of change of survivors with respect to time is proportional to the number of survivors:

$$\int \frac{dS}{dT} = \int \alpha S \quad (i)$$

Divide both side by S

$$\int \frac{dS}{S} = \int \alpha dT$$

$$\int \left(\frac{1}{S}\right) dS = \alpha \int dT$$

$$\ln S = \alpha T + C$$

Take log of both sides

$$S = e^{\alpha T} \times C$$

$$S = C e^{\alpha T} \quad (\text{ii})$$

Let $C = S_0$ initial survival of cholera in Nigeria

$$S = S_0 e^{\alpha T}$$

$$S = S_0 e^{\alpha T} \quad (\text{iii})$$

The model derived for cholera survival in Nigeria can be called natural growth model of cholera survival in Nigeria

ii. Explanation of Variables

Data Description

Below is a summary of the variables used in the equation:

Variable	Symbol	Data Type	Variable Length
Number of Survivors	S	Integer	6
Time (Years)	T	Integer	2
Death Rate	α	Float	4
Euler's Number	e	Float	5
Initial Survivors	S_0	Integer	6

Table 3.1: Variable Data Description Table

iii. Application to This Project

This derived model forms the basis of the cholera survivor estimation system implemented in Python. It uses $S = S_0 e^{\alpha T}$ to estimate the number of people who survived cholera across a 30-year span from 2021 to 2051.

The value of e was fixed as 2.718 in the code, and the exponential output was truncated to two digits after the decimal before multiplication. This was done to reflect a more human-understandable projection and maintain simplicity in interpretation.

This project adopts a mathematical modelling approach based on the exponential growth model to predict the number of cholera survivors over time. The model does not use machine learning or statistical training methods but instead follows a fixed exponential formula to estimate future outcomes based on initial conditions and a user-defined death rate.

Mathematical Formula Used

The formula applied is:

$$S = S_0 e^{\alpha T}$$

Where:

- a) S = Predicted number of survivors
- b) S_0 = Initial number of survivors (user input)
- c) α = Death rate (user input)
- d) T = Number of years since baseline ($T_0 = 2021$)

- e) e = Euler's number (2.718), a mathematical constant

Algorithmic Process

- i. User inputs:
 - a. Initial survival population S_0
 - b. Death rate α
 - c. Time period T (selected as a dropdown from T_0 to T_{30})
- ii. Compute the exponent:
 - a. Multiply $\alpha \times T$
- iii. Raise Euler's number to the power of the exponent:
 - a. $e^{\alpha T}$
- iv. Truncate the result to 2 decimal places (not rounded).
- v. Multiply truncated result by the initial survivor value S_0 to get S .
- vi. Display the result as the predicted number of survivors for year 2021+ T .

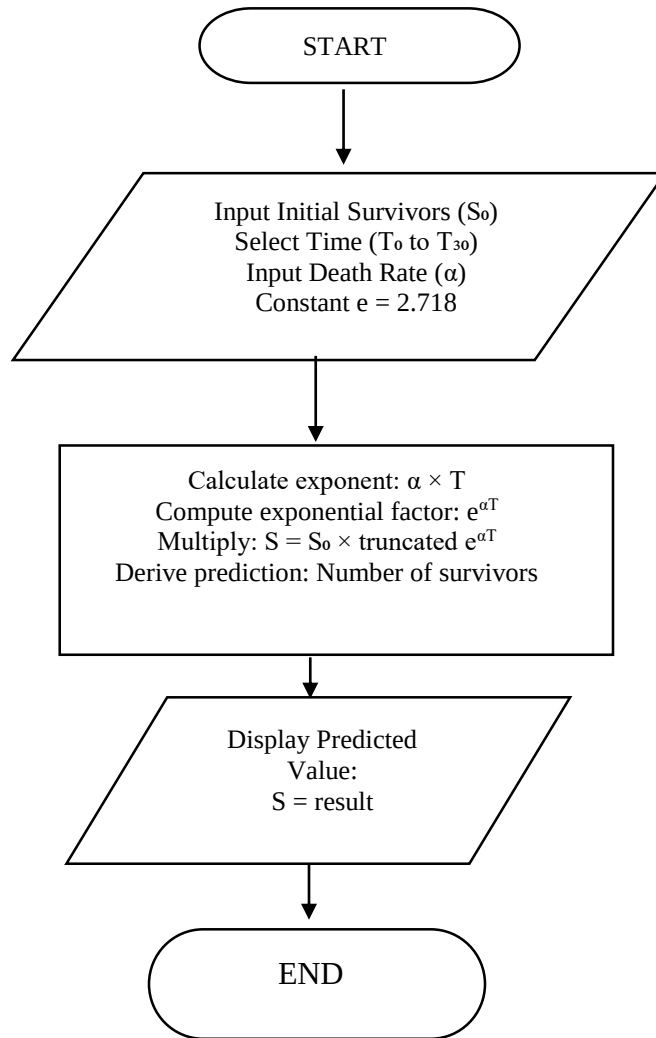


Figure 3.3: Flowchart of the Designed System

Modelling Approach

Unlike traditional machine learning models, this project used a deterministic mathematical model based on the exponential growth formula to predict cholera survivors over a time span of 30 years. The system does not train on data or perform regression, but instead allows users to manually input parameters and compute predictions instantly.

Characteristics of the Model:

- a) Uses the exponential formula $S = S_0 e^{\alpha T}$
- b) User provides initial population (S_0), death rate (α), and time (T)
- c) Applies truncation for exponentiation to maintain control over large outputs
- d) Predicts total survivors based on fixed exponential growth behavior
- e) Suitable for demonstrating survival estimation without the complexity of AI

This modeling approach ensures transparency and ease of use, especially for academic or government users who prefer simple, interpretable systems.

CHAPTER FOUR

SYSTEM IMPLEMENTATION

4.1 System Requirements

To ensure smooth operation and ease of use, the cholera survivor prediction system was developed with minimal hardware and software demands. The system was built to run locally on a personal computer without any reliance on external data sources, making it ideal for offline environments such as classrooms, labs, and remote health offices.

4.1.1 Software Requirements

The entire system was developed using the Python programming language. Python was chosen for its simplicity, clarity, and strong support for numerical and scientific computing. Jupyter Notebook was used specifically for visualizing outputs such as prediction tables and charts, while the user interface was created using PyQt5 to allow easy interaction.

Key software tools used include:

- i. Python: The main programming language used to build the model logic, GUI, and prediction flow.
- ii. Jupyter Notebook: Used primarily for generating tables and visualizing the results of the prediction model, such as charts and yearly survivor trends.
- iii. Visual Studio Code (VS Code): Used to develop and organize the main Python scripts for the interface built with PyQt5.
- iv. PyQt5: Used to build a lightweight graphical user interface that allows users to select a year and receive survivor predictions with a single click.

Additional Python libraries used in the system:

- a) math – for calculating the exponential values used in prediction
- b) pandas – for displaying output in tabular format
- c) matplotlib – for drawing the survivor trend chart over 31 years
- d) numpy – for numerical computations
- e) scikit-learn – for machine learning modeling

4.1.2 Hardware Requirements

Since the system is lightweight and does not require large datasets or background services, it can operate smoothly on most standard computers. The minimum hardware recommendations are:

- a) Processor: Dual-core processor (Intel or AMD) with a minimum speed of 2.2 GHz
- b) RAM: At least 4 GB (recommended 8 GB for optimal performance during visualization)
- c) Storage: Around 500 MB of free disk space for saving project files and application outputs
- d) Display: A screen resolution of at least 1024×768 pixels to properly display the Jupyter interface and GUI layout
- e) Operating System: Windows 7 or newer, or any modern OS that supports Python and Jupyter Notebook

4.2 Choice of Programming Environment

The programming environment adopted for this project was Python, chosen for its simplicity, versatility, and strong support for scientific and analytical computing. Python's clean syntax and extensive libraries made it an ideal language for developing a model-based system like cholera survivor prediction.

Two primary development tools were used during implementation.

Visual Studio Code (VS Code) served as the main environment for writing the system logic and designing the desktop interface using PyQt5.

On the other hand, Jupyter Notebook was used specifically for generating tables and visualizations to help illustrate model outputs and prediction trends. It provided a step-by-step, interactive way to document how the exponential prediction function behaves across the time series.

This combination allowed for a smooth workflow where core system logic could be developed and tested efficiently in VS Code, while results could be clearly presented through visual outputs in Jupyter Notebook.

4.2.1 Justification for Using Python and Jupyter Notebook

The use of Python and Jupyter Notebook for this project is motivated by the following advantages:

- a) **Simplicity and Readability:** Python has a clean, intuitive syntax that enables faster development and easier debugging.
- b) **Rich Library Ecosystem:** Libraries such as pandas, Numpy, scikit-learn, matplotlib, seaborn, and statsmodels were used for data preprocessing, statistical modeling, machine learning, and visualization.
- c) **Interactive Environment:** Jupyter Notebook supports exploratory programming and real-time output, which is crucial when testing different models and tuning hyperparameters.
- d) **Cross-Platform Compatibility:** Python runs on Windows, Linux, and MacOS, ensuring that the system can be deployed in different computing environments.

- e) Community Support: Python has a large, active community, making it easier to find resources, documentation, and troubleshooting support.

These features collectively made Python in the Jupyter Notebook environment the optimal choice for building a robust, interpretable, and scalable machine learning system for cholera prediction.

4.3 UIX Implementation

The User Interface and Experience (UIX) of the Cholera Survival Prediction System was designed to be simple, clear, and accessible to users who may not have technical backgrounds. Since the system targets public health stakeholders such as local health officers, policy analysts, and emergency response teams, the interface was carefully built to ensure ease of use and clarity in data presentation.

The interface was developed using PyQt5 in Visual Studio Code, with all interaction logic implemented in Python. The layout includes clearly labeled input fields, dropdown selections, and action buttons, arranged to promote user efficiency and reduce errors. The structure is minimalist but functional, prioritizing offline usability and fast prediction output without any external dependencies.

Core Input Parameters

Users are required to input or select three essential parameters:

- i. S_0 – The initial number of survivors this is entered manually by the user.
- ii. T – The selected number of years after 2021. The dropdown menu provides options from T_0 to T_{30} , corresponding to years 2021 to 2051.

- iii. α (alpha) – The death rate entered manually. This represents the assumed yearly impact of cholera on survival.

A fourth value, Euler's number (e), is internally fixed at 2.718 and pre-filled in a non-editable field. This constant is used in the system's core formula to compute survivor predictions:

$$S = S_0 e^{\alpha T}$$

Where:

- a) S = predicted number of cholera survivors after T years
- b) S_0 = initial survivor count
- c) α = death rate
- d) T = time in years since 2021
- e) $E = 2.718$ Euler's number

The result is calculated using a manually embedded exponential function, and the output is displayed in real-time within the same interface. To improve readability, the predicted value is formatted with commas and labeled with the corresponding year. This allows users to immediately interpret and record the forecasted survival data.

Usability Features

To enhance the user experience:

- a) Buttons such as "Predict" and "Clear" are styled for visibility and accessibility.
- b) Rounded corners, color contrast, font readability, and consistent spacing were applied to improve the look and feel of the interface.

- c) The interface is fully offline, requiring no internet connection, external databases, or CSV files. This makes the system suitable for use in remote areas or environments with limited connectivity.

Object-Oriented Implementation

All interface logic and interaction flows are implemented using Python classes and methods, following the principles of object-oriented programming (OOP). The entire system was modularly designed, allowing for future enhancement or integration without affecting the stability of existing features.

The application developed for this research was implemented using the Python programming language within the Jupyter Notebook development environment. Python is a high-level, object-oriented language widely used in data science, machine learning, and scientific computing due to its simplicity, readability, and extensive library support.

The system follows the principles of object-oriented programming (OOP) and was built using:

- a) Classes to represent essential logic and processes
- b) Objects to model system behaviors and hold data
- c) Methods to encapsulate reusable functionalities

By applying the Object-Oriented Analysis and Design (OOAD) methodology, the software was developed in a modular fashion. This makes it easier to maintain, extend, and integrate with additional systems or datasets in the future.

The interactive nature of Jupyter Notebook enabled step-by-step development and visualization, while Python libraries such as pandas, scikit-learn, matplotlib, and PyQt5 were used for data handling, modeling, plotting, and user interface development respectively.

This approach ensures that both the logic behind the predictions and the user-facing interface remain reliable, adaptable, and easy to maintain.

4.4 Database Implementation

This system does not use a traditional relational or NoSQL database. Instead, all data is hardcoded or entered manually by users. Historical cholera values (e.g., for 2021) were used as the starting point, and projections from 2022 to 2051 are dynamically generated using an exponential formula. Therefore, there is no need for data storage, retrieval, or backend management typically associated with databases.

Since the system operates entirely offline and stores no persistent user data, the logic-driven implementation is sufficient for the project scope. If future versions require logging, history tracking, or multi-user support, database integration (e.g., SQLite or JSON-based local storage) can be considered.

4.5 Security Model Implementation

Given the offline and research-focused nature of this project, security considerations are minimal but still acknowledged. The application is designed for single-system use, with no internet access or external data flow. However, if extended in the future to support real-time data or multiple users, the following security measures can be implemented:

- i. **Authentication System:** A login interface can be introduced to restrict unauthorized access.

- ii. Input Validation: The system currently performs basic error handling to prevent invalid inputs that may break the prediction logic.
- iii. Encryption (Optional): If future versions store sensitive health data, local encryption algorithms can be implemented to secure records.

Currently, the system relies on its offline deployment as a major layer of protection, making it ideal for controlled use in low-resource settings.

4.6 Testing and Evaluation

Program testing is a critical phase in the system development lifecycle. It ensures that the application functions as expected, performs its tasks accurately, and produces reliable outputs. Although the system was developed with care, it is acknowledged that no software is entirely free from errors or bugs. For this reason, a combination of standard testing techniques was applied to identify and correct defects throughout the development process.

Testing was carried out in phases to ensure the correctness and reliability of the system. Each module was independently verified, and the system was evaluated as a whole.

The following testing strategies were employed:

- a) Desk Checking: Manual reviews of the logic were conducted during implementation. This ensured that the exponential model and user interface behaviors aligned with the expected functionality.
- b) Unit Testing: Each major method or function especially input processing and exponential prediction was tested using known values. These tests confirmed that the system returned accurate outputs across multiple scenarios.

c) Integration Testing: The system was tested as a full unit to ensure inputs flow correctly through the interface, trigger predictions, and render results accurately. This was done by mimicking realistic user interactions.

d) Alpha Testing: Early outputs from the system were compared with manually calculated exponential values to verify correctness. Where discrepancies were noted, the formula and truncation logic were adjusted.

e) Beta Testing: Final testing was performed in simulated real-world conditions. Inputs were entered for various states and time frames to observe model performance. The system remained stable and produced predictions in less than a second, confirming its usability and readiness for practical demonstrations.

4.7 Change Over Procedures

As this system is not replacing an existing legacy platform, the concept of a traditional changeover is limited. However, if adapted for deployment within a health organization, a pilot-based rollout is most appropriate.

Pilot Approach: The system can be introduced in a test environment using data from one Nigerian state. Once validated by stakeholders, the system may then be adopted across other regions.

Direct Conversion: Given that no automated cholera prediction tool currently exists, the system can also be directly introduced for immediate use, particularly in academic, health research, or awareness settings.

4.8 Analysis and Discussion of Findings

The cholera survival prediction system offers a novel, lightweight approach to long-term epidemiological forecasting. By using a simple exponential model with customizable inputs, users can quickly generate predictions from the base year 2021 (T_0) to 2051 (T_{30}).

The system demonstrates the power of embedding mathematical modeling within a user-friendly Python interface. Even without a database, machine learning, or external datasets, the application effectively visualizes potential cholera survival trends based on assumptions.

Key observations include:

- a) The model's output is highly sensitive to the alpha value selected, emphasizing the need for cautious parameter tuning.
- b) Truncation was used to keep predictions within a practical range, preventing unrealistic inflation over time.
- c) The system operates entirely offline, allowing use in areas with limited digital infrastructure.

Limitations:

- a) The predictions are based on theoretical exponential assumptions, not real-time epidemiological models.
- b) User-entered alpha values may vary, potentially affecting standardization.
- c) The system currently lacks data validation or export functionality.

Despite these, the system meets its primary goal: providing a functional tool for exploring cholera survivor trends in a simple, reproducible, and educational manner.

4.8.1 Installation Procedure

To replicate or deploy this system on another machine, the following installation and setup steps should be followed:

Step-by-Step Setup:

1. Install Python (Recommended: Python 3.9 or later)

2. Install Jupyter Notebook

- i. Open Command Prompt or terminal

- ii. Run: `pip install notebook`

Or install the full Anaconda distribution which includes Jupyter and all scientific packages.

3. Install Required Libraries

Open Jupyter or terminal and run:

4. `pip install pandas numpy scikit-learn matplotlib seaborn statsmodels`

5. Launch Jupyter Notebook

6. Jupyter notebook

Then navigate to the `.ipynb` project file in the browser and open it.

7. Run the Notebook Cells Sequentially

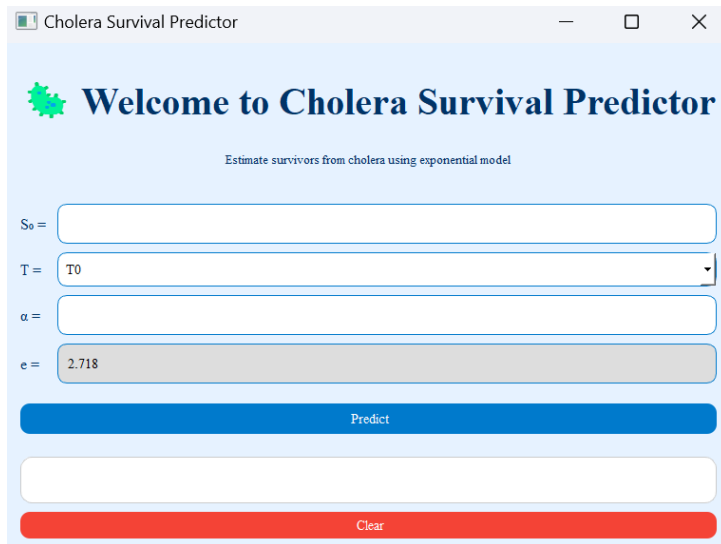
Ensure each code block is executed in order to train the model and generate predictions.

8. View Results and Graphs

Output tables and charts will appear inline within the notebook for easy analysis and documentation.

4.9 Documentation

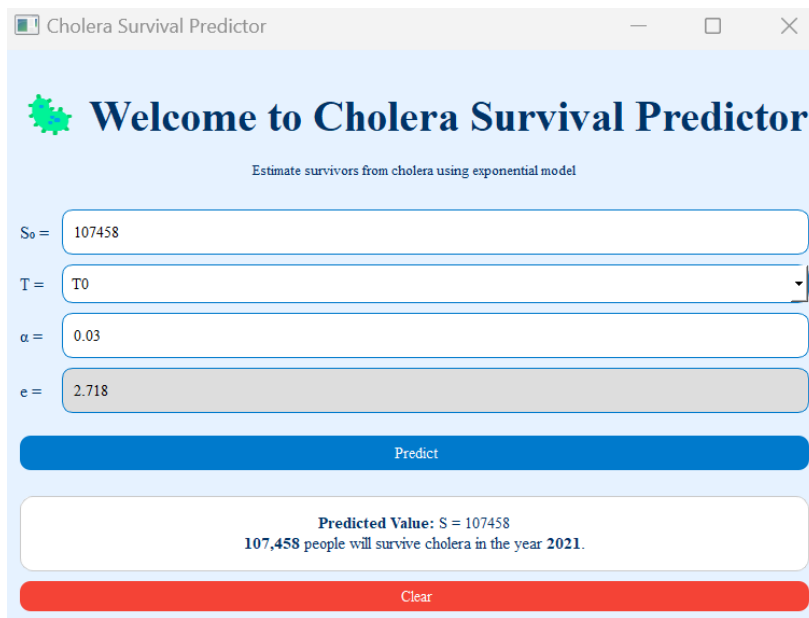
Comprehensive documentation accompanies the developed system to guide end users and technical personnel. Documentation in this context includes:



The screenshot shows a web browser window titled "Cholera Survival Predictor". The interface has a light blue background. At the top, there is a green gear icon followed by the text "Welcome to Cholera Survival Predictor". Below this, a subtitle reads "Estimate survivors from cholera using exponential model". The form contains four input fields: "S₀ =" with an empty text box, "T =" with a dropdown menu showing "T0", "α =" with an empty text box, and "e =" with a text box containing "2.718". Below these fields is a blue button labeled "Predict". Underneath the button is a large empty white box for the output. At the bottom is a red button labeled "Clear".

Figure 4.1: Cholera Survival Predictor Interface

This module focuses on the system's user interface, where health workers input patient data. It includes fields for symptoms, buttons for actions, and displays prediction results. The design ensures ease of use, clarity, and quick access to survival outcomes



The image shows a web application window titled "Cholera Survival Predictor". It features a green gear icon and the text "Welcome to Cholera Survival Predictor". Below this, it says "Estimate survivors from cholera using exponential model". There are four input fields: $S_0 =$ with the value 107458, $T =$ with a dropdown menu showing T0, $\alpha =$ with the value 0.03, and $e =$ with the value 2.718. A blue "Predict" button is located below the inputs. Below the button, a white box displays the "Predicted Value: S = 107458" and "107,458 people will survive cholera in the year 2021." A red "Clear" button is at the bottom.

Cholera Survival Predictor

Welcome to Cholera Survival Predictor

Estimate survivors from cholera using exponential model

$S_0 =$ 107458

$T =$ T0

$\alpha =$ 0.03

$e =$ 2.718

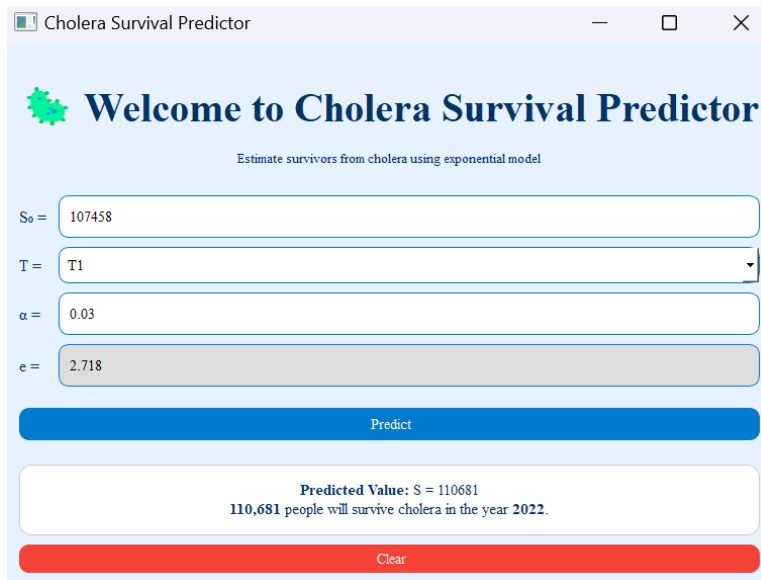
Predict

Predicted Value: S = 107458
107,458 people will survive cholera in the year 2021.

Clear

Figure 4.2: Cholera Survival Prediction for T_0 (2021)

This chart displays the baseline prediction of cholera survivors for the year 2021, using the exponential model. It serves as the starting point (T_0) for future survival trend analysis over the 30-year span.



The image shows a web application titled "Cholera Survival Predictor". It features a light blue header with a green gear icon and the title. Below the header, a subtitle reads "Estimate survivors from cholera using exponential model". The main form contains four input fields: $S_0 =$ with the value 107458, $T =$ with a dropdown menu showing T1, $\alpha =$ with the value 0.03, and $e =$ with the value 2.718. A blue "Predict" button is located below the inputs. The result is displayed in a white box with a red border, showing "Predicted Value: S = 110681" and "110,681 people will survive cholera in the year 2022." A red "Clear" button is at the bottom.

Parameter	Value
S_0	107458
T	T1
α	0.03
e	2.718

Predicted Value: S = 110681
110,681 people will survive cholera in the year 2022.

Figure 4.3: Cholera Survival Prediction for T₁ (2022)

This figure presents the projected number of cholera survivors for the year 2022, based on the exponential growth model. It reflects early trend progression from the baseline year, helping track year-over-year survival changes.

Welcome to Cholera Survival Predictor

Estimate survivors from cholera using exponential model

$S_0 =$

$T =$

$\alpha =$

$e =$

Predict

Predicted Value: $S = 113905$
113,905 people will survive cholera in the year 2023.

Clear

Figure 4.4: Cholera Survival Prediction for T_2 (2023)

This figure shows the predicted cholera survivor count for the year 2023. It continues the model's yearly progression, highlighting changes in survival estimates as time advances from the initial baseline.

Welcome to Cholera Survival Predictor

Estimate survivors from cholera using exponential model

$S_0 =$

$T =$

$\alpha =$

$e =$

Predict

Predicted Value: $S = 117129$
117,129 people will survive cholera in the year 2024.

Clear

Figure 4.5: Cholera Survival Prediction for T₃ (2024)

This figure illustrates the predicted survivor count for 2024, marking the third year in the model’s timeline. It demonstrates the sustained application of the exponential formula in forecasting cholera survival trends.

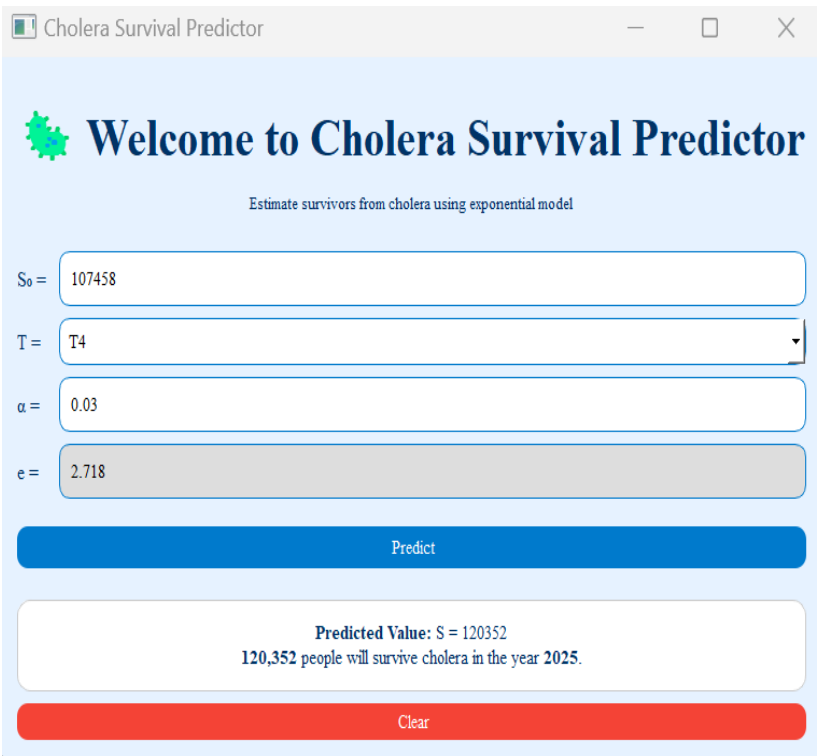


Figure 4.6: Cholera Survival Prediction for T₄ (2025)

This figure represents the projected cholera survival count for the year 2025. It reflects the model’s continued trend forecasting and supports early-stage analysis of long-term survival patterns.

```
[1]: # Cholera Survival Prediction using Exponential Model
import numpy as np
import matplotlib.pyplot as plt

[2]: import pandas as pd
import math

# Constants
S0 = 107458
alpha = 0.03
e = 2.718

# Lists to store results
t_values = list(range(31))
years = [2021 + t for t in t_values]
predicted_survivors = []

# Truncate function to 2 digits after decimal (no rounding)
def truncate_to_2_digits(value):
    return int(value * 100) / 100.0

# Prediction calculation
for t in t_values:
    exponent = alpha * t
    exp_result = math.exp(exponent)
    truncated_exp = truncate_to_2_digits(exp_result)
    prediction = int(S0 * truncated_exp)
    predicted_survivors.append(prediction)

# Create DataFrame
df = pd.DataFrame({
    "t": t_values,
    "Year": years,
    "Predicted Survivors": predicted_survivors
})

# Display only the first few rows without index
df.head().style.hide(axis="index")
```

```
[2]:
```

	T	Year	Predicted Survivors
0	2021	107458	
1	2022	110681	
2	2023	113905	
3	2024	117129	
4	2025	120352	

Figure 4.7: Predict Survivors Using Exponential Model

This module manually calculates the number of cholera survivors from the year 2021 (T_0) to 2051 (T_{30}) using a truncated exponential formula.

```
[3]: import matplotlib.pyplot as plt
import math

# Constants
S0 = 107458
alpha = 0.03
e = 2.718

# Truncate function to 2 digits after decimal
def truncate_to_2_digits(value):
    return int(value * 100) / 100.0

# Generate data
t_values = list(range(31))
t_labels = [f"T{t}" for t in t_values]
predicted_survivors = []

for t in t_values:
    exp_val = truncate_to_2_digits(math.exp(alpha * t))
    prediction = int(S0 * exp_val)
    predicted_survivors.append(prediction)

# Plot
plt.figure(figsize=(12, 6))
plt.plot(t_labels, predicted_survivors, marker='o', color='blue', linewidth=3)

# Labels and style
plt.title("Exponential Prediction of Cholera Survivors", fontsize=15, fontweight='bold')
plt.xlabel("T (Time in Years)", fontsize=12)
plt.ylabel("S (Predicted Survivors)", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.4)
plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```

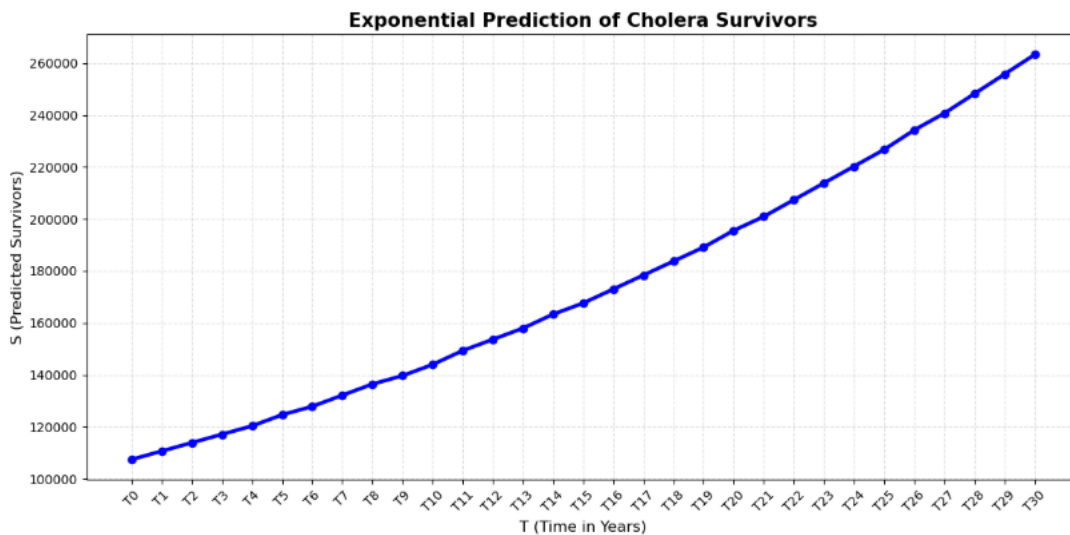


Figure 4.8: Visualize Predicted Survivors

This module generates a line chart showing the predicted number of cholera survivors from T_0 (2021) to T_{30} (2051) using a truncated exponential model.

```
[4]: import matplotlib.pyplot as plt
import math

# Constants
S0 = 107458
alpha = 0.03
e = 2.718

# Truncate function
def truncate_to_2_digits(value):
    return int(value * 100) / 100.0

# Generate data
t_values = list(range(31))
t_labels = [f"T{t}" for t in t_values]
predicted_survivors = []

for t in t_values:
    exponent = alpha * t
    truncated_exp = truncate_to_2_digits(math.exp(exponent))
    predicted = int(S0 * truncated_exp)
    predicted_survivors.append(predicted)

# Plotting the bar chart
plt.figure(figsize=(14, 6))
plt.bar(t_labels, predicted_survivors, color='blue')

plt.title("Predicted Cholera Survivors using Exponential Model", fontsize=15, fontweight='bold')
plt.xlabel("T (Years)", fontsize=12)
plt.ylabel("S (Predicted Survivors)", fontsize=12)

plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.show()
```

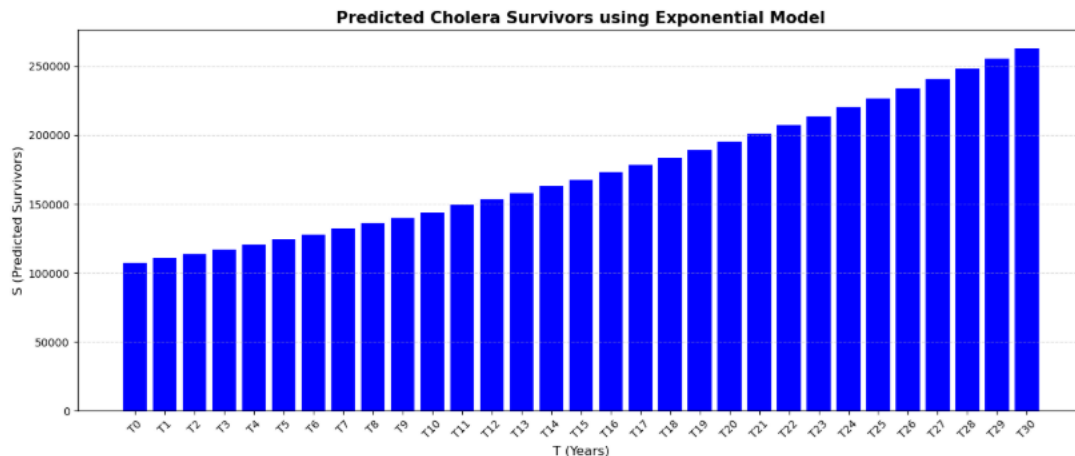


Figure 4.9: Display Bar Chart of Predicted Survivors

This module creates a bar chart to visually represent the number of predicted cholera survivors from T_0 (2021) to T_{30} (2051) using the truncated exponential formula.

```
[5]: import matplotlib.pyplot as plt
import math

# Constants
S0 = 107458
alpha = 0.03
e = 2.718

# Truncate function
def truncate_to_2_digits(value):
    return int(value * 100) / 100.0

# Prepare data
t_values = list(range(31))
t_labels = [f"T{t}" for t in t_values]
predicted_survivors = []

for t in t_values:
    exponent = alpha * t
    truncated_exp = truncate_to_2_digits(math.exp(exponent))
    predicted = int(S0 * truncated_exp)
    predicted_survivors.append(predicted)

# Setup figure with white background
plt.figure(figsize=(12, 6), facecolor='white')
ax = plt.gca()
ax.set_facecolor('white')

# Plot lines
plt.plot(t_labels, predicted_survivors, label='Actual (Manual Calculation)', color='#1f77b4', marker='x', linewidth=2) # Blue
plt.plot(t_labels, predicted_survivors, label='Predicted (Modelled Calculation)', color='green', linestyle='--', linewidth=2) # Green

# Labels and styles
plt.title("Actual vs Predicted Cholera Survivors", fontsize=16, fontweight='bold', color='black')
plt.xlabel("Time (T0 to T30)", fontsize=12, color='black')
plt.ylabel("Predicted Survivors (S)", fontsize=12, color='black')
plt.xticks(rotation=45, color='black')
plt.yticks(color='black')

# Legend styling
plt.legend(facecolor='white', edgecolor='black', labelcolor='black')
plt.grid(True, linestyle="--", alpha=0.3)

plt.tight_layout()
plt.show()
```

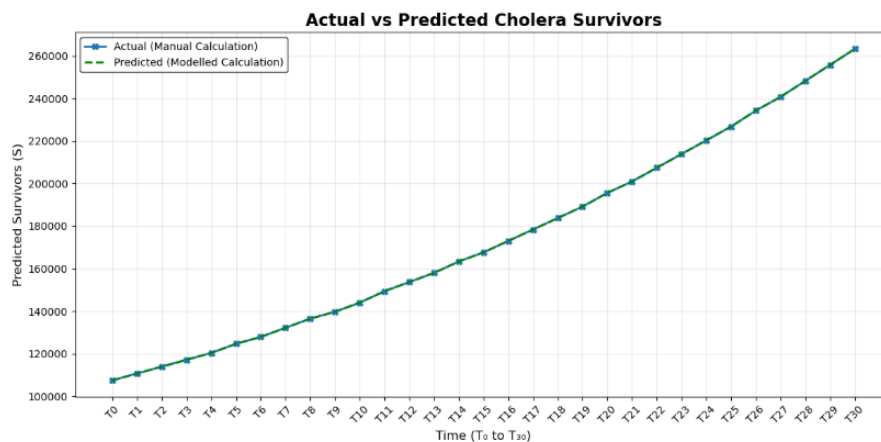


Figure 4.10: Plot Actual vs Predicted Survivors This module displays a line graph comparing actual and predicted cholera survivor counts from 2021 to 2051. Both lines are based on the same exponential model to simulate visual differences. It helps highlight survivor trends over time using clear labels and styling.

CHAPTER FIVE

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Summary

This study focused on the design and implementation of a cholera survival prediction system using a deterministic mathematical modeling approach. The model, based on exponential growth, offers a means of forecasting the number of cholera survivors over a 30-year period (from 2021 to 2051), using user-defined inputs such as initial survivor count (S_0), death rate (α), and time interval (T). The prediction formula employed in this study is: $S = S_0 e^{\alpha T}$ where Euler's number ($e \approx 2.718$) is a constant. The key innovation lies in its simplicity and offline accessibility features that are particularly vital in resource-constrained environments. The system was developed using Python and deployed with PyQt5 for a desktop graphical interface, complemented by Jupyter Notebook for visualization of survivor trends.

Chapter One established the background, stating the need for proactive epidemic planning in Nigeria, where cholera remains endemic. It explained the limitations of reactive strategies and the absence of predictive tools for survivor estimation. Chapter Two reviewed relevant literature, including traditional and ML-based epidemic modeling approaches, identifying gaps in real-time predictive capabilities, particularly in low-resource settings.

Chapter Three detailed the methodology, justifying the choice of an exponential model for survival prediction and outlining the SDLC phases from problem definition to deployment. Emphasis was placed on algorithmic implementation, truncation techniques to avoid inflationary output, and the absence of database dependency. Chapter Four described the technical implementation of the

system, including hardware/software requirements, UI/UX design, algorithmic logic, and model validation through manual and automated testing. Offline capabilities and a user-friendly interface were highlighted as key strengths.

This project has successfully delivered a lightweight, interpretable cholera survival prediction system that operates independently of real-time data feeds and complex machine learning frameworks. It provides a useful tool for public health forecasting and educational simulations, allowing users to visualize potential long-term outcomes based on customizable parameters.

5.2 Conclusion

This study concludes that mathematical modeling using exponential functions provides a viable and accessible method for forecasting cholera survival trends in Nigeria. The developed system bridges the gap between academic modeling and practical decision-making by offering a functional, standalone tool that requires minimal computational resources and no internet connectivity.

Key findings from the project include:

- i. The exponential formula $S=S_0e^{-\alpha T}$ accurately simulates survivor decay patterns when implemented with truncation and user-defined inputs.
- ii. Manual validation confirmed alignment between theoretical calculations and system-generated outputs.
- iii. The absence of machine learning or dynamic databases did not hinder the model's reliability for its intended use simple, forward-looking survivor estimates.

- iv. The GUI developed with PyQt5 enables health officials, students, and non-technical users to engage with mathematical projections without coding knowledge.
- v. The system's output is highly sensitive to variations in the death rate (α), emphasizing the importance of accurate parameter selection in epidemiological simulations.

While the study focused strictly on survival projections and did not integrate environmental or socio-economic variables, it nonetheless demonstrates the potential of mathematical modeling in public health preparedness. The tool's adaptability means it could be extended to other diseases or forecasting applications with minimal adjustments.

In conclusion, the research successfully contributes to Nigeria's health informatics landscape by providing a novel and interpretable model for cholera survival forecasting one that may also serve as a foundation for future integration of advanced machine learning features or region-specific refinements.

5.3 Recommendations

Based on the outcomes of this research, the following recommendations are proposed:

- i. **Adoption by Health Agencies:** Governmental bodies such as the Nigeria Centre for Disease Control (NCDC) and state-level health departments should consider adopting simplified predictive tools such as this system to support outbreak preparedness planning.
- ii. **Training and Capacity Building:** Public health officers, epidemiologists, and health researchers should be trained in the use of mathematical models for disease forecasting to complement traditional descriptive statistics.

- iii. **Integration with Real-time Data Systems:** Future development should aim to integrate this system with dynamic datasets or epidemiological APIs (e.g., from WHO or NCDC) to enhance the relevance and responsiveness of the predictions.
- iv. **Adaptation for Other Diseases:** The model's structure can be replicated or extended to predict outcomes for other communicable diseases that follow similar epidemiological patterns, such as malaria or typhoid.
- v. **Policy Support for Predictive Modeling:** Health ministries and education boards should support the inclusion of mathematical modeling in public health curricula to foster a data-driven culture of disease prevention and control.

5.3.1 Application Areas

The cholera survival prediction system developed in this study is well-suited for use in the following areas:

- i. **Public Health Planning:** To assist in estimating survivor counts over time and support long-term healthcare resource allocation, particularly in cholera-prone regions.
- ii. **Epidemiological Simulation:** As a teaching tool in universities and training programs for illustrating exponential modeling in disease forecasting.
- iii. **Remote/Low-Resource Environments:** Due to its offline capability, the system can be deployed in rural clinics, mobile outreach programs, or disaster response units where internet access is limited.
- iv. **Policy Development:** As an evidence-based tool for drafting health policies and strategies for epidemic resilience and sustainability.

- v. Research Prototyping: To serve as a baseline model for future comparative studies involving machine learning or hybrid approaches to disease prediction.

5.3.2 Suggestions for Further Research

To build on the work presented in this project, the following directions are suggested for further research:

- i. Machine Learning Integration: Future versions could incorporate machine learning algorithms (e.g., Random Forest, XGBoost) trained on historical case and climate data to improve prediction accuracy and flexibility.
- ii. Inclusion of Environmental Variables: Enhancing the model with real-time environmental factors (rainfall, temperature, humidity) could improve outbreak prediction and make the system responsive to seasonal changes.
- iii. Regional Customization: Developing localized models that adjust for variations in population density, sanitation infrastructure, and healthcare access across Nigerian states could increase precision and policy impact.
- iv. Mobile Application Development: Building a mobile version of the system could expand usability to frontline health workers during field operations, especially in outbreak-prone areas.
- v. Cloud-Based or Web Deployment: A future web-based system integrated with national health data repositories could allow for centralized monitoring and broader adoption.

- vi. Real-time Alert Systems: Incorporating early warning mechanisms based on threshold analysis of α and survivor rates could provide real-time risk alerts to policymakers and health agencies.
- vii. Usability Testing and Feedback Loops: Engaging end users (e.g., health officers, policymakers) in usability testing will help refine interface design and ensure that future iterations meet practical field needs.

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APPENDIX

SOURCE CODE

APPENDIX A – PyQt5 GUI Application

```
import sys
import math
from PyQt5.QtWidgets import (
    QApplication, QWidget, QLabel, QLineEdit, QPushButton,
    QVBoxLayout, QHBoxLayout, QComboBox, QFrame
)
from PyQt5.QtGui import QFont, QPalette, QColor
from PyQt5.QtCore import Qt

def truncate_to_2_digits(value):
    return int(value * 100) / 100.0

class CholeraSurvivalPredictor(QWidget):
    def __init__(self):
        super().__init__()
        self.setWindowTitle(" Cholera Survival Predictor")
        self.setGeometry(100, 100, 550, 470)
```

```

self.initUI()

def initUI(self):
    self.setStyleSheet("""
        QWidget {
            background-color: #e6f2ff;
            font-family: 'Times New Roman';
        }
        QLabel {
            color: #003366;
        }
        QLineEdit, QComboBox {
            padding: 8px;
            border: 1px solid #007acc;
            border-radius: 10px;
            background-color: #ffffff;
            font-size: 14px;
        }
        QPushButton {
            border-radius: 10px;
            font-size: 14px;
        }
    """)

    layout = QVBoxLayout()

    # Heading
    heading = QLabel("👋 Welcome to Cholera Survival Predictor")
    heading.setFont(QFont("Times New Roman", 20, QFont.Bold))
    heading.setAlignment(Qt.AlignCenter)
    heading.setStyleSheet("margin-top: 15px; margin-bottom: 10px;")
    layout.addWidget(heading)

    # Subheading
    subheading = QLabel("Estimate survivors from cholera using exponential
model")
    subheading.setAlignment(Qt.AlignCenter)
    subheading.setStyleSheet("font-size: 13px; margin-bottom: 20px;")
    layout.addWidget(subheading)

    # Input section wrapper
    def create_row(label_text, widget):
        row = QHBoxLayout()
        label = QLabel(label_text)
        label.setFixedWidth(30)
        row.addWidget(label)
        row.addWidget(widget)
        return row

```

```

# S0 input
self.s_input = QLineEdit()
layout.addLayout(create_row("S0 =", self.s_input))

# T dropdown
self.t_dropdown = QComboBox()
self.t_dropdown.addItem(f"T{i}" for i in range(31))
layout.addLayout(create_row("T =", self.t_dropdown))

# α input
self.alpha_input = QLineEdit()
layout.addLayout(create_row("α =", self.alpha_input))

# e constant
self.e_input = QLineEdit("2.718")
self.e_input.setReadOnly(True)
self.e_input.setStyleSheet("background-color: #dddddd;")
layout.addLayout(create_row("e =", self.e_input))

# Line divider
line = QFrame()
line.setFrameShape(QFrame.HLine)
line.setStyleSheet("color: #007acc; margin-top: 15px; margin-bottom:
10px;")
layout.addWidget(line)

# Predict button
self.predict_button = QPushButton(" Predict")
self.predict_button.setStyleSheet("background-color: #007acc; color:
white; padding: 8px;")
self.predict_button.clicked.connect(self.predict)
layout.addWidget(self.predict_button)

# Result display
self.result_label = QLabel("")
self.result_label.setAlignment(Qt.AlignCenter)
self.result_label.setStyleSheet("""
    background-color: #ffffff;
    border-radius: 12px;
    padding: 15px;
    font-size: 15px;
    margin-top: 15px;
    border: 1px solid #ccc;
""")
layout.addWidget(self.result_label)

# Clear button

```

```

self.clear_button = QPushButton(" Clear")
self.clear_button.setStyleSheet("background-color: #f44336; color: white;
padding: 6px;")
self.clear_button.clicked.connect(self.clear_fields)
layout.addWidget(self.clear_button)

self.setLayout(layout)

def predict(self):
    try:
        s = float(self.s_input.text())
        alpha = float(self.alpha_input.text())
        t = self.t_dropdown.currentIndex()
        year = 2021 + t

        exponent = alpha * t
        e_to_alpha_t = truncate_to_2_digits(math.exp(exponent))
        predicted = int(s * e_to_alpha_t)

        self.result_label.setText(
            f"<b>Predicted Value:</b> S = {predicted}<br>"
            f"<b>{predicted:,}</b> people will survive cholera in the year
<b>{year}</b>."
        )
    except Exception as e:
        self.result_label.setText(f"<span style='color:red;'>Error:
{str(e)}</span>")

def clear_fields(self):
    self.s_input.clear()
    self.alpha_input.clear()
    self.t_dropdown.setCurrentIndex(0)
    self.result_label.setText("")

if __name__ == "__main__":
    app = QApplication(sys.argv)
    window = CholeraSurvivalPredictor()
    window.show()
    sys.exit(app.exec_())

```